Freight Rate Volatility: Air vs Road in the EU

Final Project Report - Six-State Analysis, Dashboard and Procurement Decision Tool

Parag Khare | May 2026 cohort

8/14/26

Data Analyst, Liora (DataScientest)

Freight Rate Volatility: Air vs Road in the EU

Final Project Report - Six-State Analysis, Dashboard and Procurement Decision Tool

Parag Khare | Data Analyst training, Liora (DataScientest), May 2026 cohort | August 2026

Table of Contents

1. Executive summary and reader's guide

This report answers one business question with evidence: which mode, air or road, shows the more significant freight rate volatility in the EU, and what seasonal pattern drives it. The answer is air. On average, EU air freight rates are about 1.63 times as volatile as road by coefficient of variation, the gap is statistically significant, and both modes lift into the second quarter, air far more strongly than road. The pattern is an average rather than a universal rule, because road carries the larger swing in the Netherlands and Poland.

The study compares six member states: Germany (the primary market), Italy, Spain, the Netherlands, Poland, and France. France is included through the national air-freight series published by INSEE, the French statistics office, because Eurostat carries no French air index. Adding France strengthened the headline rather than weakening it: the air-over-road volatility ratio widened from 1.5 to 1.63, and France turned out to be the second most volatile air lane of the six, after Italy.

The work was built for a working audience: a category, pricing, or procurement manager who decides how to contract each mode, when to run a tender, and how wide a freight budget should be. It draws on twelve years of freight pricing, tendering, and procurement experience and adds a reproducible quantitative layer on top.

The report has three parts. Part A establishes the evidence: the price indices, the data sources including the French addition, a clean and documented data pipeline, the full set of visualizations each confirmed by a statistical test, and a country-by-country read. Part B turns that evidence into an interactive Power BI dashboard, the Rate Risk Monitor, and explains the design and the reasoning behind it. Part C covers the final-report material: the difficulties encountered, the main contribution, the results against an industry benchmark, a goal-by-goal review, the overall conclusion, and the avenues for continuation.

The headline results are summarised below and developed through the report.

Question	Result	Evidence
Which mode is more volatile	Air, about 1.63x road by CV	Mean CV 0.155 vs 0.095
Is the gap real	Yes, highly significant	Levene 65.0, p about 3e-15
Are the modes seasonal	Both, air much more so	Kruskal-Wallis air p about 1e-10, road p about 5e-4
When do rates peak	The second quarter	Air +3.2% mean quarter-on-quarter in Q2
Where does France sit	Second most volatile air lane	Air CV 0.186 vs road 0.074
Would the timing rule have paid	Yes, in 75% of market-years	Backtest across 85 market-years, mean edge 1.8%
Is the gap a pandemic artifact	No	CV ratio 1.35 ex-COVID, 1.41 pre-COVID, gap significant on every window
Does the pattern still hold	The Q2 climb does	10 of 12 market-years out of sample, 2024-2025

Table 1. Headline results at a glance (six states).

New in this version: chapter 8 validates the recommendations before Part C reviews the project. The tender-timing rule is backtested year by year, the volatility gap is re-tested without the pandemic years, and the seasonal pattern is checked out of sample on air quarters the analysis never touched.

One caveat to carry from the start: French air is measured at the air-freight level (INSEE CPF 51.21), while the other five use the broader Eurostat air-transport division (NACE H51), which includes passenger air. A freight-only series tends to swing more, so part of France's high air ranking reflects the purer definition, not only a real-world gap. The report states this openly wherever France appears.

2. Introduction

2.1 Context of the project

This project sits at the intersection of two things I know and one I am building. I bring twelve years in freight pricing, tendering, and procurement, on both the buyer and the seller side, and I am completing the Data Analyst training to add a quantitative layer to that experience. The project applies the analytical skills of the course to a problem I understand commercially, rather than to a generic dataset. It is both my course submission and the portfolio piece I will use to move into a pricing, category, or procurement role.

From a technical point of view, the project compares two families of official Services Producer Price Index (SPPI) series: air transport (NACE H51) and road freight transport (NACE H49.4, coded H494 in the data). Both are quarterly index series, rebased so the 2021 average equals 100, and both are published not seasonally adjusted (NSA). Five of the six countries draw both series from Eurostat. France draws its air series from INSEE, the French national statistics office that compiles the same kind of producer price index for France, because Eurostat publishes no French air series. The SPPI measures the price that service providers charge for the service, so it is a clean, neutral proxy for freight rate behaviour over time, free of any single carrier or lane bias.

From an economic point of view, freight rate volatility is a direct cost-management problem. Rate movement decides when a buyer should fix a contract rate, when to run a tender, how long a contract should run, and how much budget contingency to hold. Transport is the largest single component of logistics cost and freight can reach around a tenth of the landed cost of a product, so a few percentage points of rate movement is real money on a sizeable freight budget. The air market makes the point. During the 2024 Red Sea disruption, global air spot rates climbed sharply, with some Asia-to-Europe lanes up by large double digits year on year, while European road rates moved only a few index points over the same period. A mode whose price can jump in weeks must be contracted and budgeted differently from one that grinds slowly, and that difference is what this project measures.

Air and road are the two modes a European shipper actively manages on the same lanes. Sea is slower and priced separately, and rail is a minor share on most corridors, so the air-versus-road choice is where pricing and mode-shift decisions are made week to week. They are substitutes at the margin for time-sensitive freight, and their rate behaviour shapes the real trade-off between speed and cost.

2.2 Objectives and success criteria

The central objective is to answer one business question with evidence: which mode shows the more significant rate volatility, and what seasonal pattern drives it. Around that sit three supporting objectives: to document and justify a clean, defensible data pipeline, to characterise each series through visualization and descriptive statistics, and to back every visual claim with an appropriate statistical test so the conclusions carry into the dashboard.

On expertise, this is an individual project, so all of the work is mine. On the business side I am the domain expert. Twelve years pricing and tendering air and road freight means I can read the series against real market events, judge whether a result is plausible, and translate a statistical finding into a procurement decision. Rather than a single external interview, I cross-checked the direction of the findings against published industry intelligence, including the Transport Intelligence and Upply European road freight benchmark, which characterises road rates as slow-moving, and air-cargo commentary that describes air as event-driven. The finding here, that air is the more volatile mode, is consistent with that industry view.

The success criteria follow the methodology. The analysis is judged on a clean and justified pipeline, a clear characterisation of each series, and statistical tests that confirm the visual claims. The dashboard is judged on a defensible design built to named evaluation criteria. The final report is judged on the combined story, the difficulties log, the contribution, and the goal-by-goal review. The sections below map to those criteria directly.

2.3 Scope and structure of this report

Scope is deliberately tight. The project uses two price indices only, air and road, and excludes volume and global context layers, so the analysis stays deep rather than broad and every figure is defensible. The comparison runs on the member states with usable data in both modes. This keeps the question sharp: a like-for-like read of rate volatility and seasonality between two EU freight modes, now across six markets.

Part A covers the data and the analysis: where the data comes from, including the French INSEE addition, what it measures, which variables matter, every cleaning and feature step with its justification, then the distributions, levels, relationships, the two core questions of volatility and seasonality each backed by a test, and a country-by-country read of all six markets. Part B covers the dashboard: the visualization problem, the goals and evaluation criteria, the design and interactivity of the Power BI Rate Risk Monitor as built and re-pointed to six states, and how the analysis shaped it. Part C covers the difficulties, the contribution and results, the conclusion, and the continuation, followed by a bibliography and an appendix that documents the data, the measures, the limits, and each code file.

Part A. Data and analysis

3. Understanding and manipulation of the data

3.1 Framework

The Eurostat datasets are free open data, downloaded with no API key and no paywall, so the work is fully reproducible. The owner and publisher is Eurostat, the statistical office of the European Union, and both Eurostat series come from the quarterly services producer price collection, dataflow sts_sepp_q. The French air series is free open data from INSEE, the French national statistics office, through its BDM database. Table 2 summarises the three sources by volume and span as downloaded and verified.

Dataset	Source	NACE / CPF	Frequency	Base	Rows	Non-null	Geos	Span
Air SPPI	Eurostat	H51	Quarterly	2021=100, NSA	3286	2066	34	1998-Q1 to 2026-Q1
Road SPPI	Eurostat	H494	Quarterly	2021=100, NSA	2247	2107	21	2003-Q1 to 2024-Q1
Air freight (France)	INSEE	CPF 51.21	Quarterly	2021=100, NSA	81	81	1	2006-Q1 to 2026-Q1

Table 2. Dataset overview as downloaded and verified.

The Eurostat files were obtained directly from the dissemination service, which returns each series as a CSV without any account or API key. The French air series is INSEE BDM series 010766683. The analysis runs on a fixed snapshot, so the figures quoted here are stable and reproducible, and the same queries can be rerun each quarter to refresh the series. Eurostat and INSEE both revise recent quarters from time to time, so a fixed snapshot is the correct basis for a report that cites exact numbers.

Each Eurostat file carries the same columns: indic_bt (the price indicator), nace_r2 (the activity code), s_adj (adjustment), unit (the index base), geo (country or aggregate), TIME_PERIOD (the quarter, for example 2015-Q1), OBS_VALUE (the index level), and OBS_FLAG (a Eurostat data-quality flag). The air file covers 34 geographies including the EU27 and euro-area aggregates. The road file covers 21 individual member states and, importantly, no EU or euro-area aggregate.

A Services Producer Price Index tracks the change over time in the prices providers charge business customers for a defined activity, here air transport and road freight transport. It is an index, not a euro rate: the 2021 average is set to 100, so a value of 112 means prices are 12 percent above their 2021 average. Using an official index rather than a single carrier quote gives a neutral, comparable measure of rate behaviour across countries and years, which is what a volatility study needs.

Two definitional choices follow. The series is kept not seasonally adjusted on purpose, because the project measures seasonality, and a seasonally adjusted series would have that signal removed. Of the two Eurostat indicator codes present, PRC_PRR (the standard producer price) is kept rather than PRC_PRR_B2B (business-to-business only), so one consistent definition applies across the five Eurostat countries. The INSEE French air series is already base 2021 and not seasonally adjusted, so it aligns to the same definition. Eurostat data-quality flags in OBS_FLAG, for example b for a break in series, e for estimated, and p for provisional, are noted where a flagged value falls inside the analysis window.

3.2 The France question and the INSEE source

France deserves its own subsection because the project's scope changed when it was added, and a careful reader will want the full provenance.

France was first excluded. Eurostat publishes no French air SPPI: the French air series in the Eurostat air file exists as a placeholder of 168 rows in which every value is null. With no air data in the Eurostat source, France could not enter an air-versus-road comparison, so the first version of the study ran on five states. The exclusion was found by a per-country non-null scan rather than assumed, which avoided a silent error in which France would have dropped out of any air calculation while still appearing in a six-country set.

The French air-freight price index does exist, however, at the national source Eurostat would otherwise aggregate. INSEE compiles a producer price index for French air freight transport under CPF code 51.21, published in its BDM database as series 010766683, base 2021, données brutes (not seasonally adjusted), running 2006-Q1 to 2026-Q1. This is the live base-2021 replacement for an older base-2015 series. The French air series was located at INSEE rather than on the open data.gouv.fr portal, whose API returned nothing for the query. French road, by contrast, is already in Eurostat (H494), so it was kept there like the other five.

Two checks support mixing the sources. First, the road values agree across sources: Eurostat French road at 2024-Q1 reads 114.5 against INSEE road at 114.7, within rounding, so the two statistical offices measure the same road market. Second, INSEE is the national source Eurostat would itself aggregate for a European figure, so the French air series is the same kind of official producer price index as the Eurostat series, not a private benchmark. France is restored as a sixth state, with air from INSEE and road from Eurostat.

The one honest caveat to carry is a granularity difference. The five Eurostat air series are NACE H51, the whole air-transport division, which includes passenger air. The French air series is INSEE CPF 51.21, air freight specifically, one level deeper and freight-only. A freight-only series tends to swing more than a series diluted by steadier passenger traffic, so part of France's high air ranking reflects the purer freight definition rather than only a real-world gap. France is the second most volatile air lane of the six, and that ranking should be read with the definition in mind. The report states this caveat wherever France appears, and Section 4.7.6 reads the French series with it in view.

3.3 Relevance

The variables that matter for the objective are TIME_PERIOD, geo, and OBS_VALUE. TIME_PERIOD gives the time axis, geo lets the comparison run at country level, and OBS_VALUE is the index level from which every metric is derived. The other columns are filter keys: indic_bt, nace_r2, s_adj, and unit select one consistent definition of the series across countries.

There is no single supervised target variable in the modelling sense, because this is a descriptive and dashboarding project rather than a prediction project. The target concept is the quarterly rate change and its dispersion, in other words the volatility of each mode. The features derived from the level to express that concept are the within-series z-score, the rolling four-quarter standard deviation, the coefficient of variation, the quarter-on-quarter percentage change, and the calendar quarter.

The data has real limits, which I state up front because they shape the method. First, the Eurostat air index (NACE H51) includes passenger air transport, so it is a broad air-rate proxy rather than a cargo-only price. France is the exception, because its air series is INSEE CPF 51.21, air freight only, which is purer but sits one level below the H51 used for the other five, so France should be read with that granularity difference in mind. Second, road has no EU or euro-area aggregate, so a clean air-versus-road comparison can only be made at member-state level. Third, each country starts in a different year, so the two modes must be aligned to a common window before they are compared. Figure 1 shows how the comparison windows differ in length across the six markets.

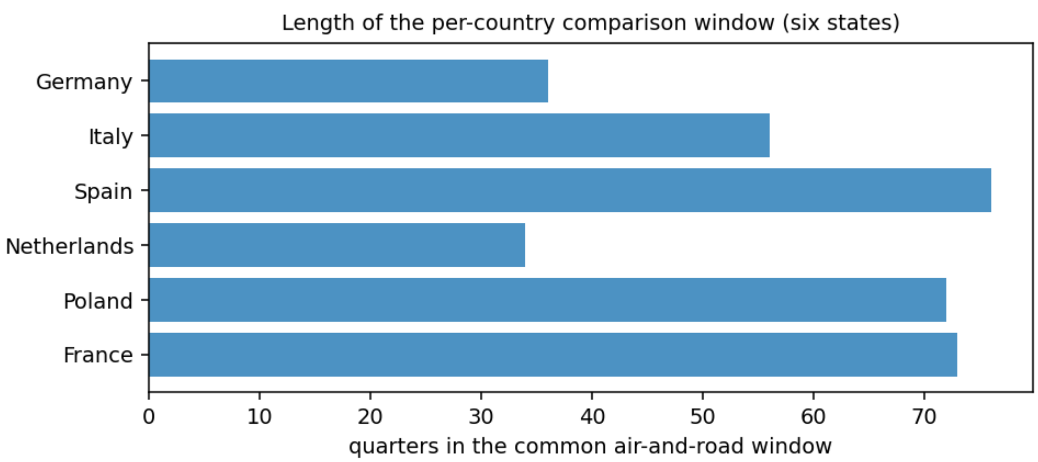

Figure 1. Length of the per-country comparison window. France and the longer-running markets carry more history than Germany and the Netherlands.

3.4 Pre-processing and feature engineering

Every cleaning step is recorded with a one-line justification, both for transparency and because each choice is a likely defence question.

- Filter to indic_bt = PRC_PRR, unit = I21, and s_adj = NSA. Reason: PRC_PRR is the standard producer price relative, I21 fixes the 2021=100 base, and NSA keeps the seasonality, which is exactly what the project measures. The INSEE French air series is already base 2021 and not seasonally adjusted, so it aligns to the same definition.
- Keep the six common member states DE, IT, ES, NL, PL, FR. Reason: these six have usable air and road data once France is sourced from INSEE for air and Eurostat for road.
- Parse TIME_PERIOD (for example 2015-Q1) into a quarter-end date. Reason: a real datetime allows sorting, rolling windows, and resampling.
- Reshape both series into one tidy long table with columns date, geo, mode, and value. Reason: long format feeds both the statistical tests and the dashboard slicers.
- Drop leading nulls per series and do not interpolate. Reason: filling gaps invents low-variance points and would understate volatility, the very thing being measured.
- Restrict each country to the window where both modes exist, ending 2024-Q1, the last road quarter. Reason: an apples-to-apples volatility comparison needs the same window for both modes.

The per-country common windows are listed below. The combined comparison table holds 696 observations across the six countries and two modes.

Country	Common window (both modes)	Quarters
Germany (DE)	2015-Q1 to 2023-Q4	36
Italy (IT)	2010-Q1 to 2023-Q4	56
Spain (ES)	2005-Q1 to 2023-Q4	76
Netherlands (NL)	2015-Q1 to 2023-Q4	34
Poland (PL)	2006-Q1 to 2023-Q4	72
France (FR)	2006-Q1 to 2024-Q1	73

Table 3. Per-country common windows used for the air-versus-road comparison.

On normalization and standardization, the index levels are not directly comparable across countries because each is rebased to its own 2021 value and sits at a different level. To compare volatility fairly I standardized each series to a z-score within its own country and mode, that is, value minus the series mean divided by its standard deviation, giving a mean of zero and a standard deviation of one. Standardization puts every series on a common scale so the volatility comparison is not distorted by level. The coefficient of variation is computed on the raw levels, because it already normalizes by dividing the standard deviation by the mean, so no prior scaling is needed there.

3.5 Methods in depth

This subsection sets out the reasoning behind each method, because the methodology rewards a clear account of why a technique was chosen and the defence will probe it.

Standardisation with the z-score. Each country's index is rebased to its own 2021 average, so the raw levels are not comparable across countries: a value of 100 means something different in each series. The z-score subtracts each series mean and divides by its standard deviation, producing a series with mean zero and standard deviation one. On that common scale a movement of one means one standard deviation for that series, so the rolling volatility of one country can be compared with another without the level distorting the picture. The coefficient of variation needs no prior standardisation, because dividing the standard deviation by the mean already removes the level.

Coefficient of variation versus standard deviation. The plain standard deviation is in index points and grows with the level, so a high-level series can look more volatile simply because it sits higher. The coefficient of variation divides by the mean, so it measures dispersion relative to level and is fair when means differ. The report uses the coefficient of variation for the cross-country ranking and the z-scored rolling standard deviation for how volatility moves through time. Reporting both a relative measure and a time path avoids the trap of a single number.

The rolling four-quarter window. Volatility is not constant, so a single coefficient of variation hides when the risk arrived. A rolling standard deviation over four quarters, one year of quarterly data, tracks how variable the series has been recently. Four quarters is the natural cycle for quarterly data and keeps enough windows on the shorter series. An eight-quarter window would smooth away the annual signal and shorten the usable history.

The variance test. The question is whether air and road quarter-on-quarter changes differ in variance, not in average. The Levene test compares group variances, and its median-centred form, Brown-Forsythe, is robust to non-normal data. Freight rate changes have heavy tails, so the classical F-test for variances, which assumes normality, would be unreliable here. The null hypothesis is that air and road changes have equal variance, and the test rejects it at p about 3e-15, so the variance gap is not chance. A p-value is the probability of seeing a gap this large if the two modes truly had equal variance, so a tiny value makes chance an implausible explanation. It does not state the size of the effect, which is reported separately as the change-spread ratio of about 3.4 times.

The seasonality test. The question is whether the quarter-on-quarter change differs across the four calendar quarters. Kruskal-Wallis is the non-parametric one-way test, the rank-based counterpart of one-way ANOVA. It does not assume normal, equal-variance groups, which suits rate-change data with outliers. The null hypothesis is that the change distribution is the same across quarters, and the test rejects it for both modes, far more strongly for air. Working in changes rather than levels matters, because two trending levels can look similar for spurious reasons, while changes isolate the quarter-to-quarter movement a buyer actually feels.

Why no interpolation. Several series carry leading gaps. Filling them would invent smooth, low-variance points and pull the measured volatility down, biasing the very quantity the project measures. Dropping leading nulls loses a little history but keeps the variance honest, which matters more in a volatility study.

4. Visualizations and statistics

This part characterises the data through visualization and confirms each visual claim with a statistical test. It moves from distributions and outliers, through levels and relationships, to the two questions the project exists to answer, then closes with a country-by-country read of all six markets.

4.1 Results at a glance

The headline results, repeated here as the spine of the analysis, are air is the more volatile mode (about 1.63x road by CV), the gap is highly significant (Levene p about 3e-15), both modes are seasonal with air much stronger (Kruskal-Wallis air p about 1e-10), both peak in the second quarter (air +3.2% mean quarter-on-quarter), and France is the second most volatile air lane after Italy. Each is developed and tested below.

4.2 Distributions and outliers, before and after processing

Figure 2 shows the distribution of the raw rate levels for each mode. Because each series is rebased to 2021, both centre near 100, but the spread differs. Figure 3 shows the distribution after standardization, where both modes sit on a common z-score axis, confirming that standardization makes the two comparable. Figure 4 shows the distribution of quarter-on-quarter changes, the distribution that matters for volatility: the air box is visibly wider with larger tails, the first visual sign that air moves more sharply quarter to quarter.

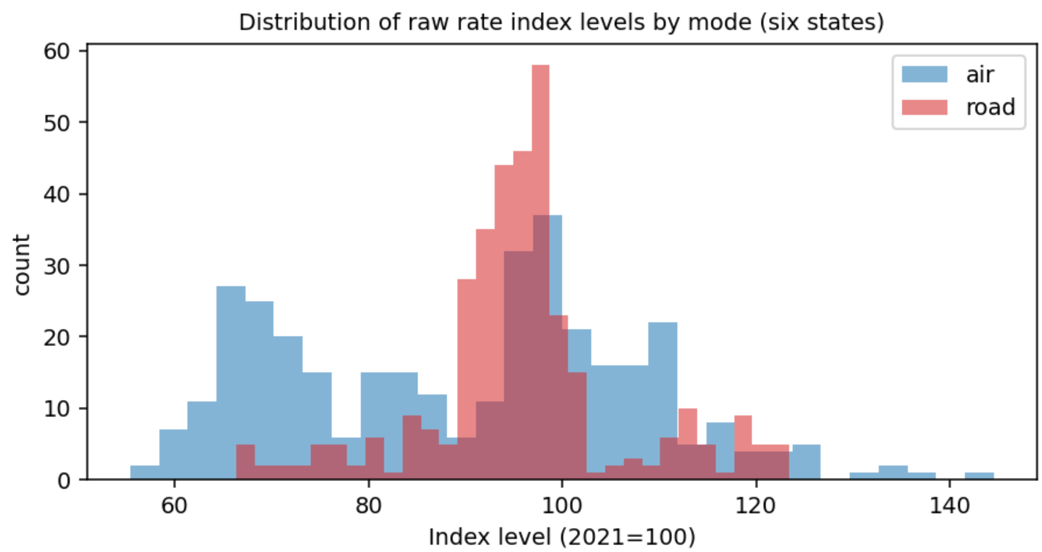

Figure 2. Distribution of raw rate index levels by mode, six states (pre-processing).

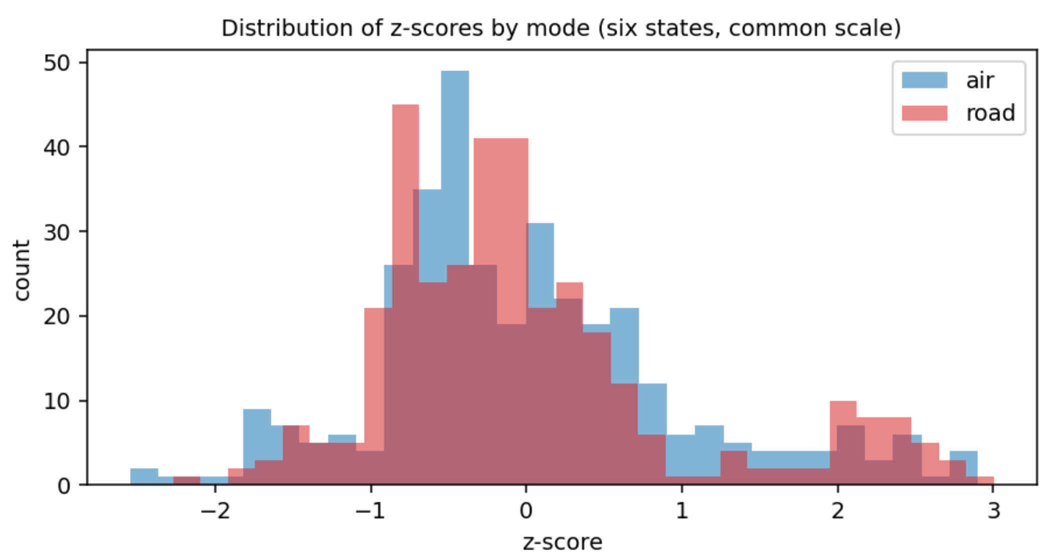

Figure 3. Distribution of z-scores by mode, six states (post standardization), on a common scale.

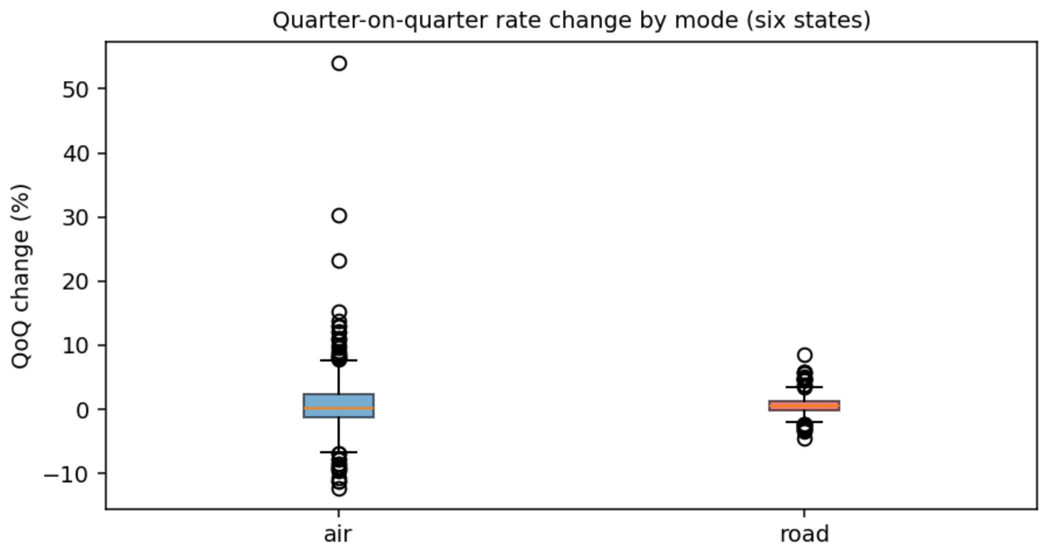

Figure 4. Quarter-on-quarter rate change by mode, six states. The air spread and outliers are larger than road.

Two things stand out. The raw level distributions are not directly comparable because each country is rebased to its own 2021 value, which is exactly why standardization is needed before any cross-series volatility claim. And the change distribution, not the level distribution, is where the modes separate: the wider air box and its longer whiskers are the visual form of the variance result that the Levene test confirms in Section 4.5. Adding France widens the air tails, consistent with the freight-only French series being the spikiest of the six.

4.3 Levels and trends

Figure 5 plots the EU27 air index over its full history of 80 quarters. It is shown for air only, because road has no EU aggregate, and it gives useful context: the long run is broadly stable until the sharp climb through 2021 and 2022, which matches the known air-cargo capacity crunch, before easing. Figure 6 places German air against German road on their common window, where the air line swings more widely around its trend than road. The full set of per-country level charts is presented in the country-by-country read in Section 4.7.

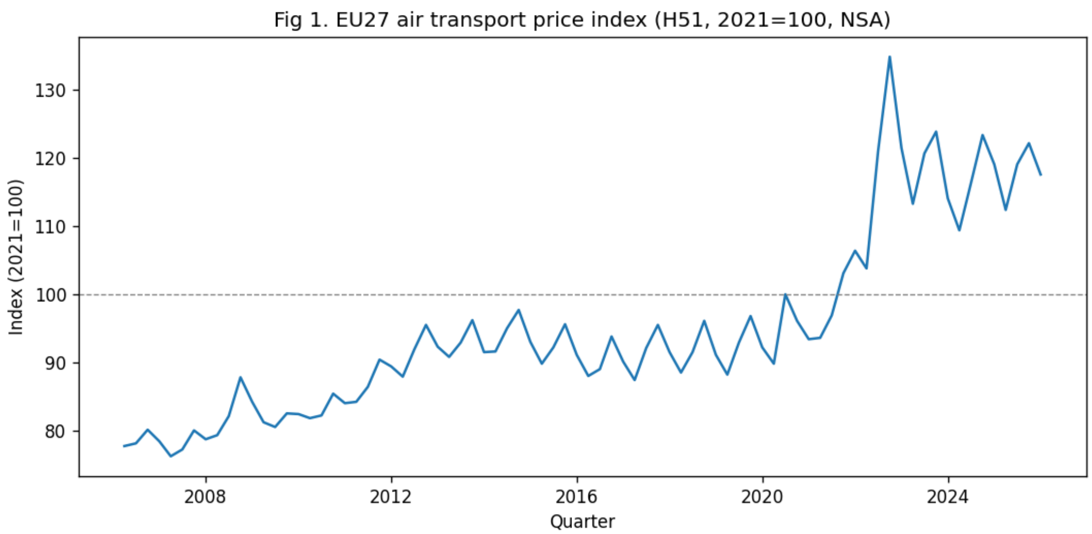

Figure 5. EU27 air transport price index, full history. The 2021 to 2022 spike matches the air-cargo capacity crunch.

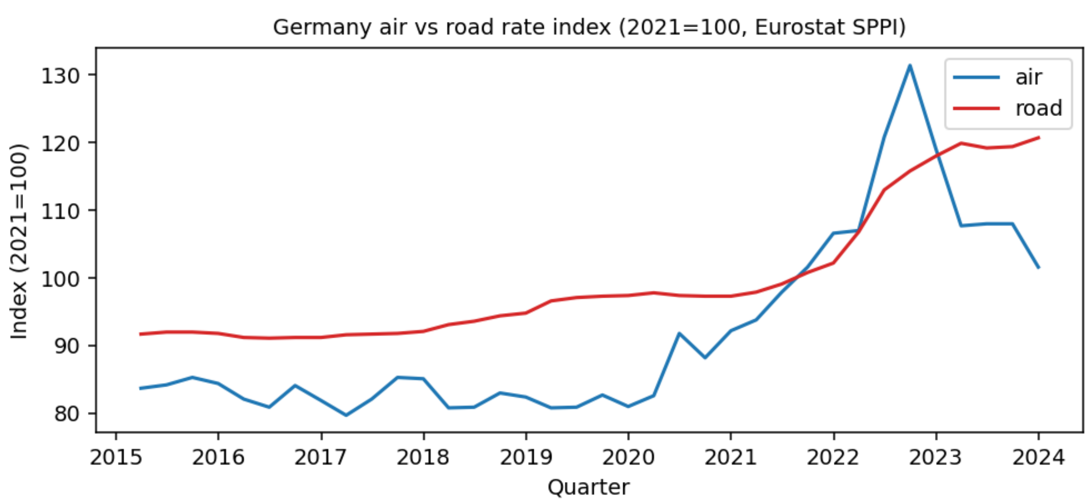

Figure 6. Germany air versus road price index over the common window.

4.4 Relationships between variables

Figure 7 shows the correlation of the air rate across the six countries. The national air series are positively correlated, which is expected because they respond to common European and global air-cargo conditions, while still differing enough that a country-level view adds information over a single line. France correlates less tightly with the others, which is consistent with its freight-only definition picking up movements the broader passenger-inclusive series smooth over. This supports analysing the markets together but reporting them by country.

Practically, the positive cross-country correlation in air means a shock tends to hit the European air market broadly rather than one country in isolation, so a buyer cannot fully diversify air rate risk by switching country. Road is more locally driven. This is one more reason the contracting response to air volatility is structural, through indexation and contract tenor, rather than geographic.

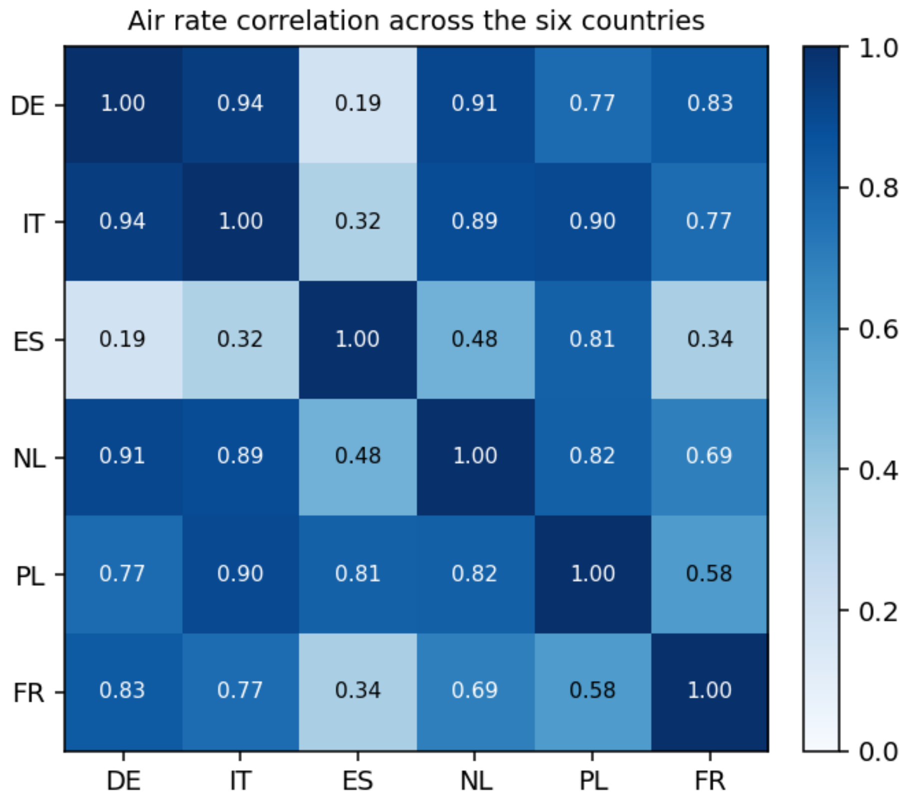

Figure 7. Correlation of the air rate index across the six countries.

4.5 Volatility: which mode moves more

Two metrics quantify volatility. The coefficient of variation (CV) measures dispersion relative to the level, and the rolling four-quarter standard deviation of the z-score shows how volatility evolves over time. Table 4 reports the CV by country and mode.

Country	Air CV	Road CV	Road minus Air
Germany	0.148	0.099	-0.049
Italy	0.282	0.053	-0.229
Spain	0.108	0.079	-0.029
Netherlands	0.086	0.099	+0.013
Poland	0.120	0.167	+0.047
France	0.186	0.074	-0.112
Mean	0.155	0.095	-0.060

Table 4. Coefficient of variation by country and mode (raw levels).

On average air is the more volatile mode, with a mean CV of 0.155 against 0.095 for road, about 1.63 times. The gap is widest in Italy, and France is the second widest air lane. Honesty matters here, and the pattern is not uniform. Air clearly leads in Germany, Italy, Spain, and France, while road is marginally more variable in the Netherlands and Poland. The headline is therefore an average and a strong Italy, France, and Germany result, not a universal rule, and the report says so. Figures 8 and 9 show the rolling volatility and the CV ranking.

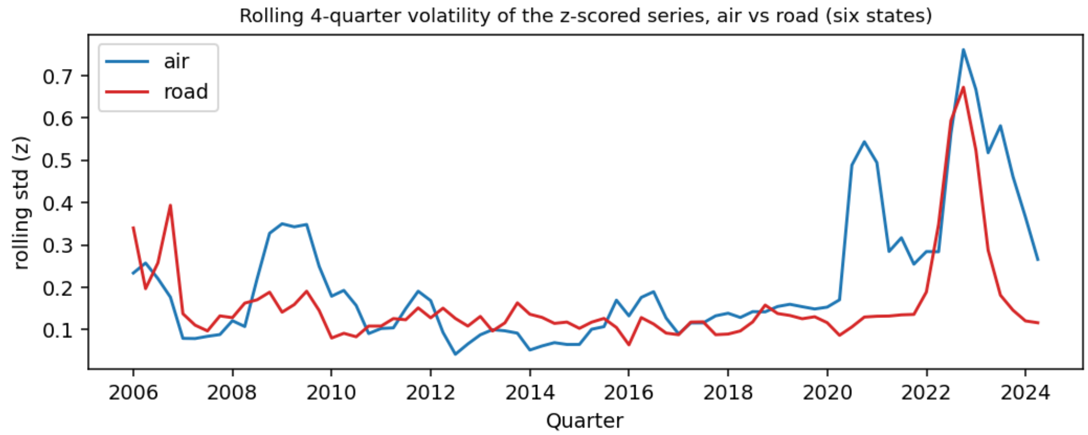

Figure 8. Rolling four-quarter volatility of the z-scored series, air versus road, averaged across the six states.

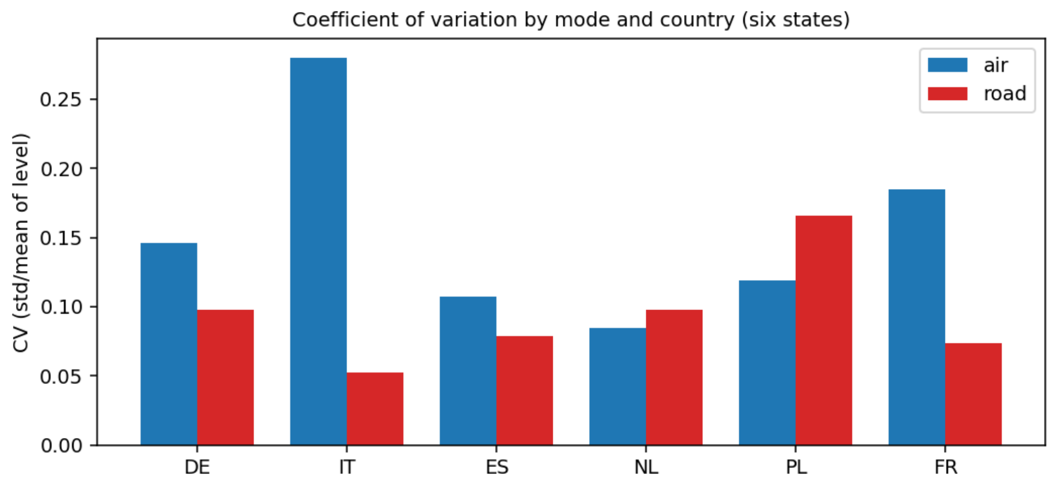

Figure 9. Coefficient of variation by mode and country, six states.

To confirm the gap is real and not chance, I ran a Levene test (Brown-Forsythe, centred on the median, robust to non-normal data) on the quarter-on-quarter changes, air against road, pooled across the six countries. The null hypothesis is that the two modes have equal variance. The test gives a statistic of 65.0 with a p-value of about 3e-15, so the null is rejected decisively. The standard deviation of air changes is 0.053 against 0.016 for road, so air changes are about 3.4 times as variable. Repeated on Germany alone, the test gives a statistic of 20.4, p about 3e-5, in the same direction. Adding France strengthened the pooled result: the statistic rose from 53.8 to 65.0 and the air-to-road change-spread ratio widened from about 2.6 to about 3.4.

Test	Scope	Statistic	p-value	Conclusion
Levene (variance)	Air vs road, 6 states	65.0	~3e-15	Reject equal variance, air more volatile
Levene (variance)	Air vs road, Germany	20.4	~3e-5	Reject equal variance, air more volatile

Table 5. Variance tests on quarter-on-quarter rate changes.

Reading the rolling volatility by country adds nuance to the average. In Germany, Italy, and France the air line sits clearly above road for most of the window, and Italy is the extreme case, with an air CV of 0.282 against a road CV of 0.053. In the Netherlands and Poland the two modes are close and road edges ahead, which is why the headline is framed as an average rather than a universal rule. A buyer should segment by lane and country, not assume one mode is always riskier.

Outlier episodes line up with known events, a useful credibility check. The largest air moves cluster in 2021 and 2022, the air-cargo capacity crunch when passenger belly capacity was withdrawn and rates climbed steeply, with a further disturbance around 2020. The French freight-only series shows the sharpest swing of all, a surge through 2020 to 2022 and a fast retreat after. Road shows nothing of comparable size in the same window. That the statistical outliers map onto real market history is a sign the series and the method behave sensibly.

4.6 Seasonality: what drives the movement

To test seasonality I grouped the quarter-on-quarter change by calendar quarter (Q1 to Q4) and applied a Kruskal-Wallis test, the non-parametric one-way test, which does not assume normal, equal-variance groups. The null hypothesis is that the change distribution is the same across quarters. For air the statistic is 49.6 with a p-value of about 1e-10. For road it is 17.6 with a p-value of about 5e-4. Both reject the null, so both modes are seasonal, but air far more strongly. Both peak in the second quarter: the mean air change is plus 3.2 percent in Q2 against plus 1.0 percent for road, while air falls back in Q4. Figure 10 shows the quarter effect and Figure 11 shows a classical decomposition of the German air series.

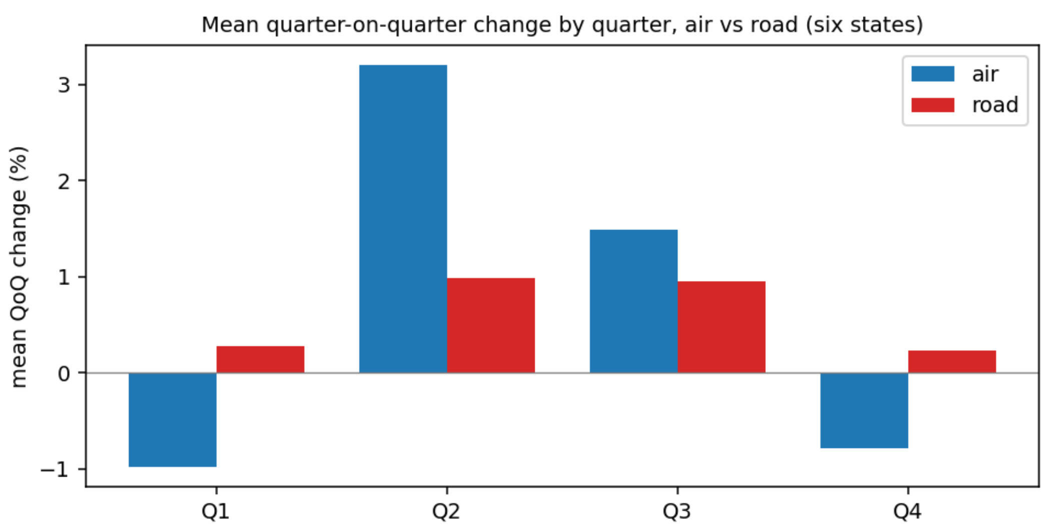

Figure 10. Mean quarter-on-quarter rate change by quarter, air versus road, six states.

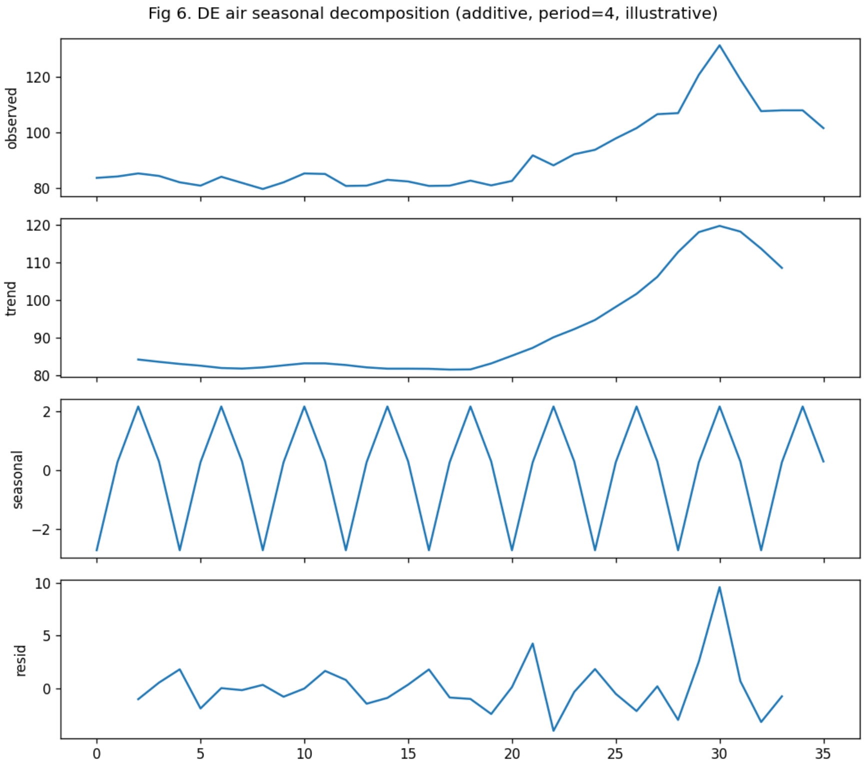

Figure 11. Seasonal decomposition of the German air series (additive, period four, illustrative).

Test	Mode	Statistic	p-value	Peak quarter
Kruskal-Wallis	Air	49.6	~1e-10	Q2 (+3.2%)
Kruskal-Wallis	Road	17.6	~5e-4	Q2 (+1.0%)

Table 6. Quarter-effect tests on rate changes.

The air Kruskal-Wallis statistic fell from 74.8 in the five-state study to 49.6 with France added. This is expected and worth owning: the French freight-only air series adds quarterly noise to the pooled pattern, which lowers the test statistic, yet the quarter effect stays highly significant and the peak stays Q2. The seasonal signal is robust to the source change, it is simply measured against a noisier sixth series.

The seasonal peak is consistent across the markets, which strengthens it as a tendering signal. Air rises into the second quarter in every one of the six countries, with the strongest mid-year lift in the markets that are also the most volatile. Road shows the same calendar shape but a much flatter one. For a buyer this means the timing rule travels: across these markets, the low season for locking an air rate sits at the turn of the year, not in the spring.

4.7 Country-by-country read

The average answers the headline, but a buyer acts on a specific market. This section reads each of the six countries in turn, pairing its air-versus-road level chart with its rolling volatility, its quarter pattern, its coefficient of variation, and the action it implies. The coefficient of variation for each country is from Table 4 and the second-quarter means are computed on the same data.

4.7.1 Germany

Germany is the primary market, given the Frankfurt focus of the target roles. German air runs at a coefficient of variation of 0.148 against 0.099 for road, so air is clearly the riskier leg, about one and a half times road by CV. Figure 6 above shows the German pair: air swings more widely around its trend, with the sharp 2021 to 2022 climb, while road grinds up steadily. The German comparison window is one of the shortest at 36 quarters, 2015 to 2023, because German road data starts later, so the German read is the most recent and the least diluted by older history. Figure 6a shows air rolling volatility above road for most of the window, and Figure 6b shows the mid-year lift: German air rises about 3.5 percent on average in the second quarter against about 0.8 percent for road. The action is to index-link or shorten German air contracts and time the air tender to the turn-of-year low season, while holding German road on longer fixed terms.

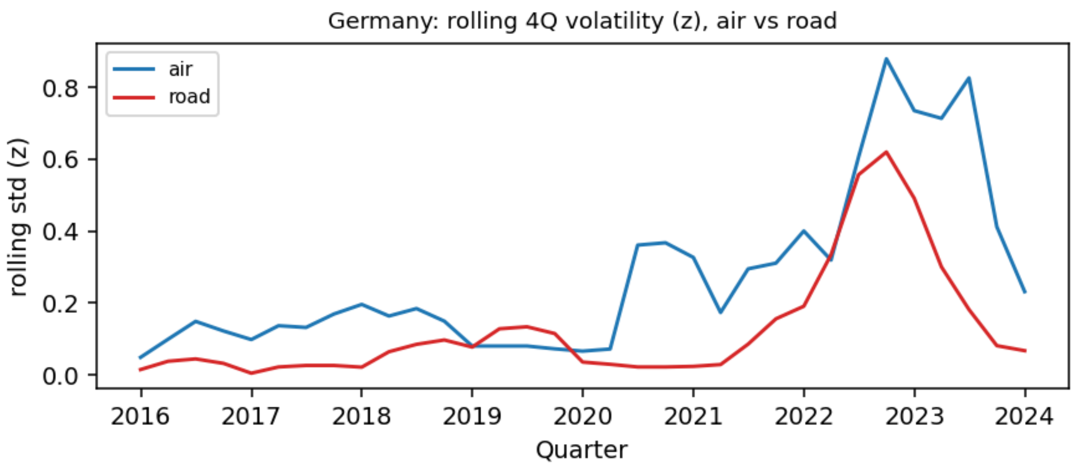

Figure 6a. Germany: rolling four-quarter volatility of the z-scored series, air versus road.

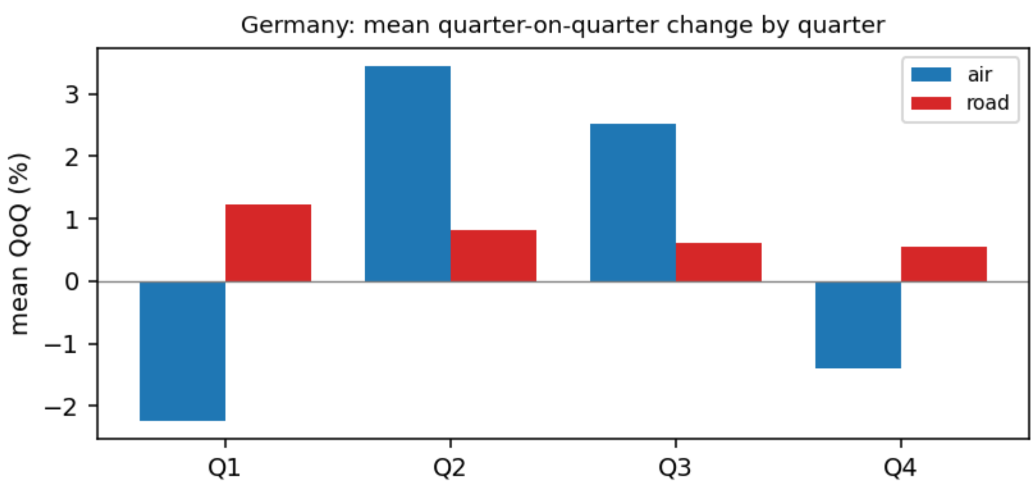

Figure 6b. Germany: mean quarter-on-quarter change by quarter, air versus road.

4.7.2 Italy

Italy is the extreme case and the widest air-to-road gap of the six. Italian air runs at a coefficient of variation of 0.282, nearly twice the six-state air average, against just 0.053 for road, the lowest road figure in the set. Figure 12 shows why: Italian air moves in large steps while Italian road is almost flat.

The mid-year lift is the strongest of the six, with Italian air up about 5.3 percent on average in the second quarter against about 0.8 percent for road in Figure 12b, and Figure 12a shows air volatility well above road across the window. For a buyer Italy is where air rate risk is the priority and where a larger risk buffer, a shorter air tenor, or a tested road or sea-air alternative earns its place. Italian road, by contrast, is safe to fix on a long term.

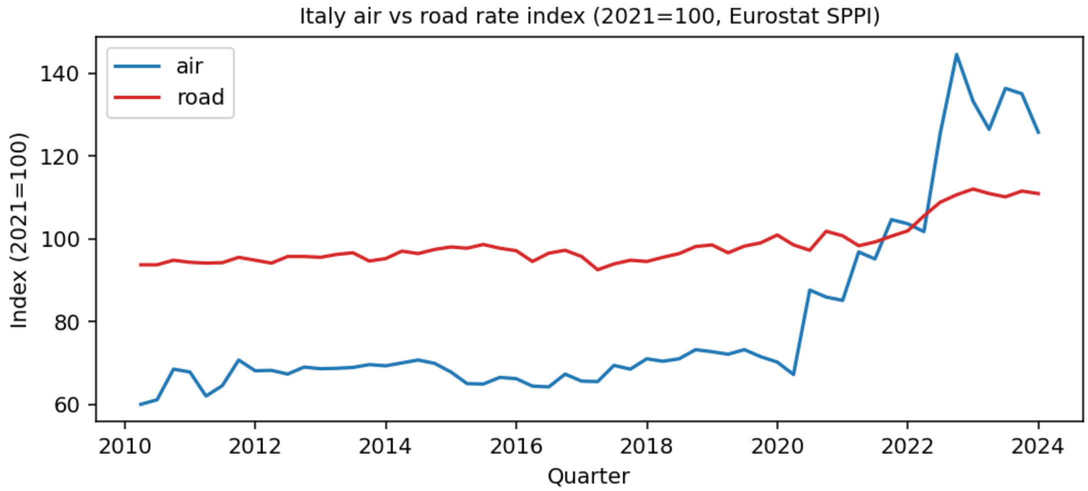

Figure 12. Italy air versus road price index over the common window.

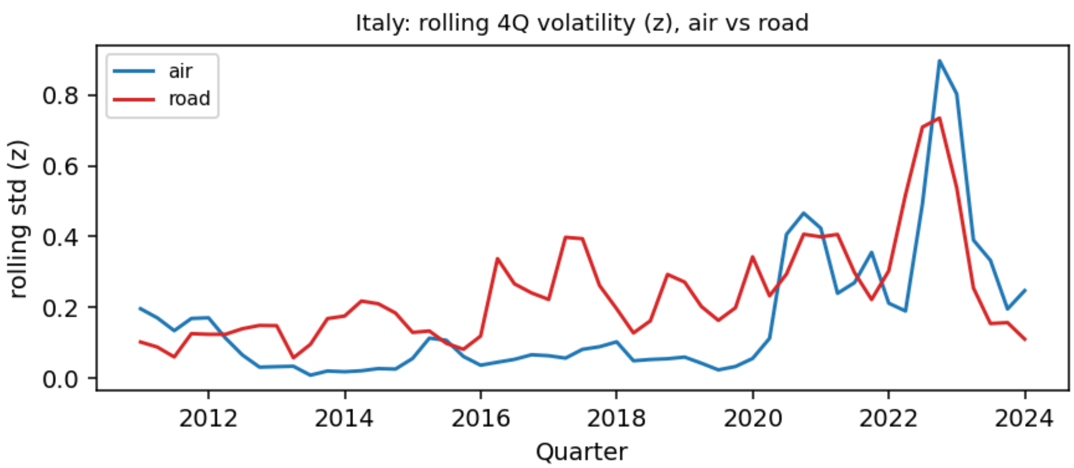

Figure 12a. Italy: rolling four-quarter volatility of the z-scored series, air versus road.

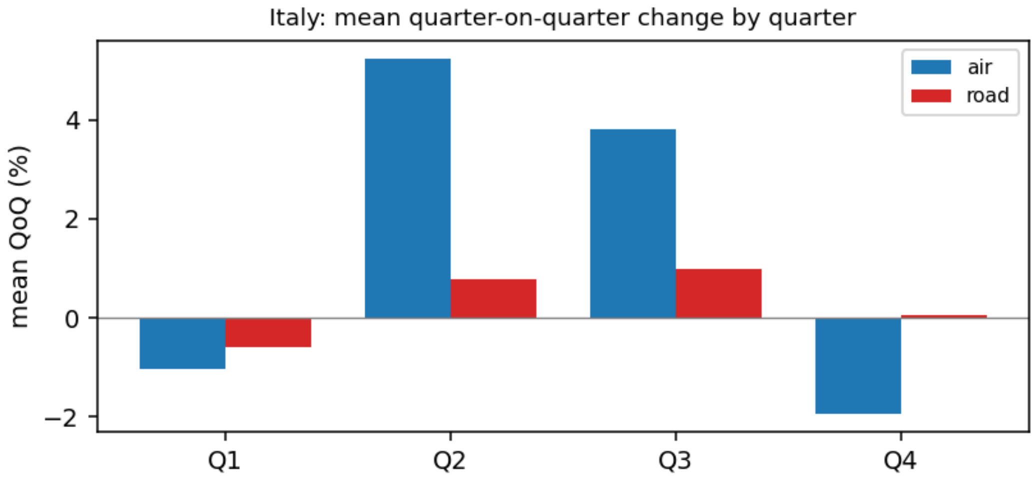

Figure 12b. Italy: mean quarter-on-quarter change by quarter, air versus road.

4.7.3 Spain

Spain shows air above road but by a moderate margin, a coefficient of variation of 0.108 for air against 0.079 for road. Figure 13 shows two series that track more closely than in Germany or Italy, with air still the more variable. Spain has the longest comparison window of the six at 76 quarters, 2005 to 2023, so the Spanish read rests on the most history. Figure 13b has Spanish air up about 2.3 percent in the second quarter against about 1.6 percent for road, the most active road of the set in the mid-year and the smallest air-road seasonal gap, and Figure 13a shows the two volatility lines running close. The action is the standard one but with a smaller buffer: index or shorten air, fix road, and tender air in the low season, sized to a moderate rather than an extreme gap.

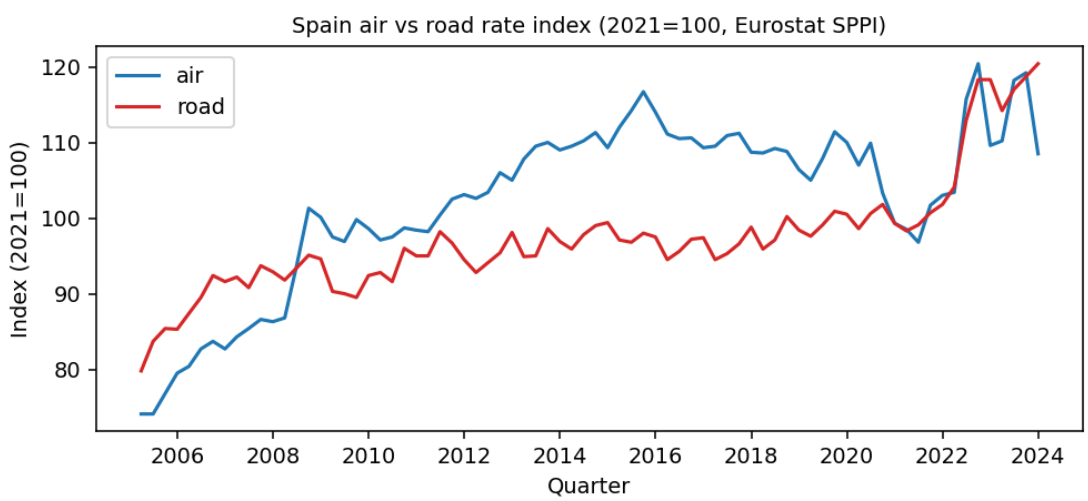

Figure 13. Spain air versus road price index over the common window.

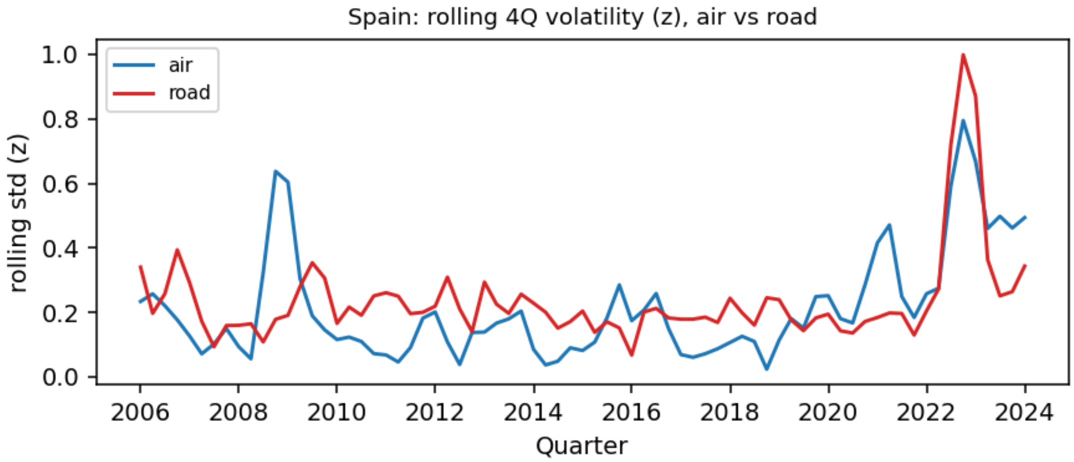

Figure 13a. Spain: rolling four-quarter volatility of the z-scored series, air versus road.

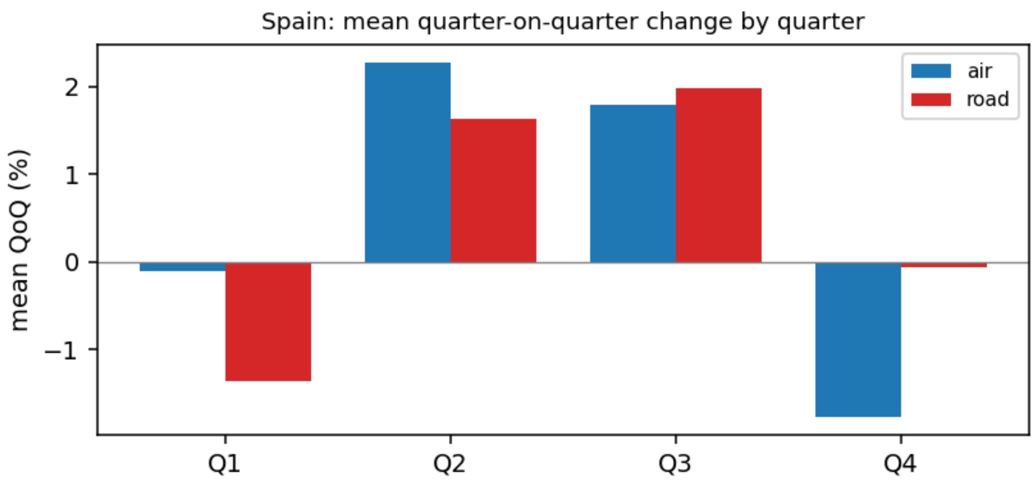

Figure 13b. Spain: mean quarter-on-quarter change by quarter, air versus road.

4.7.4 Netherlands

The Netherlands is one of the two markets that run counter to the headline, and a useful reminder that level risk and timing risk are different. Dutch air sits at a coefficient of variation of 0.086 against 0.099 for road, so road is marginally the riskier leg, and Figure 14 shows two close series with road edging ahead on variability. Yet Figure 14b shows a clear mid-year lift, Dutch air up about 4.1 percent in the second quarter against about 0.7 percent for road, so even where the overall air band is modest, the tender-timing rule still applies. Figure 14a shows road holding close to air on rolling volatility. The Dutch window is the shortest of the six at 34 quarters. The practical reading is to treat the two modes case by case in the Netherlands and to watch road as much as air when setting Dutch contract terms.

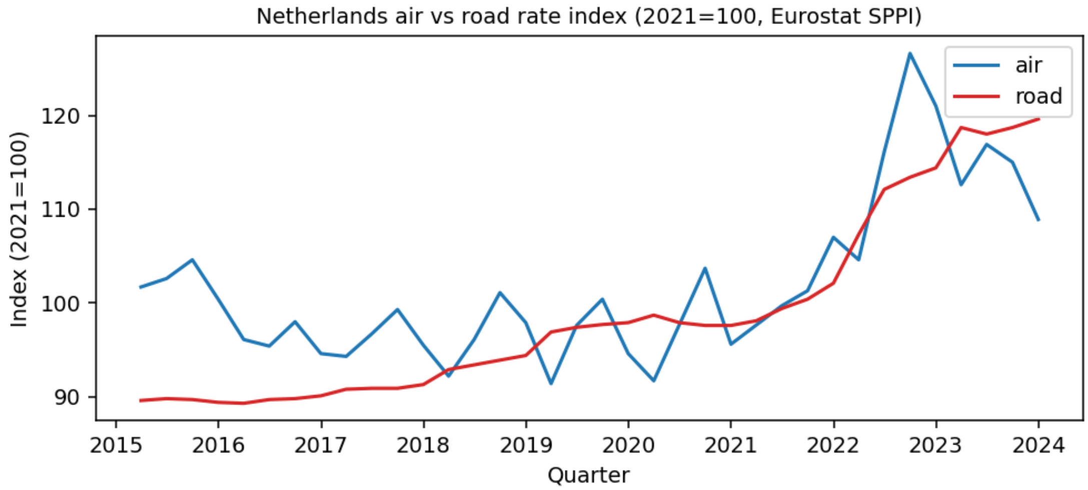

Figure 14. Netherlands air versus road price index over the common window.

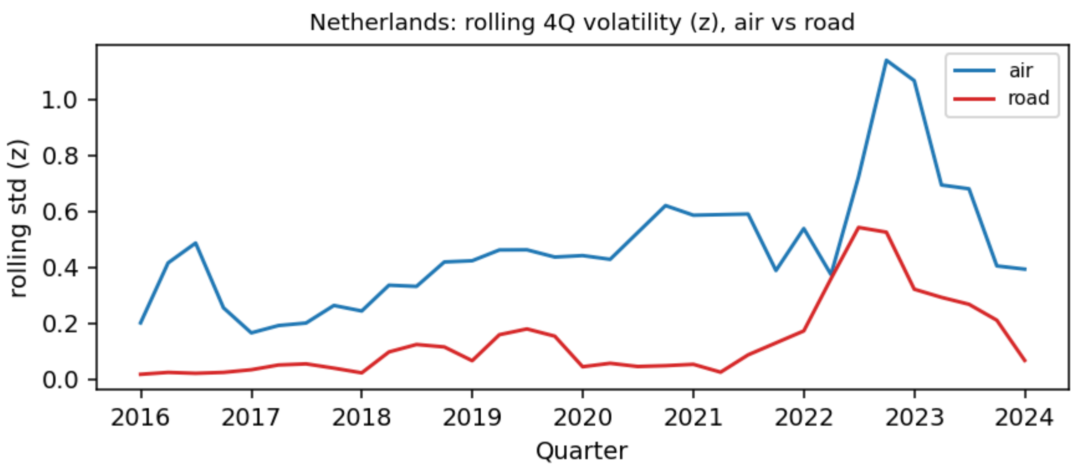

Figure 14a. Netherlands: rolling four-quarter volatility of the z-scored series, air versus road.

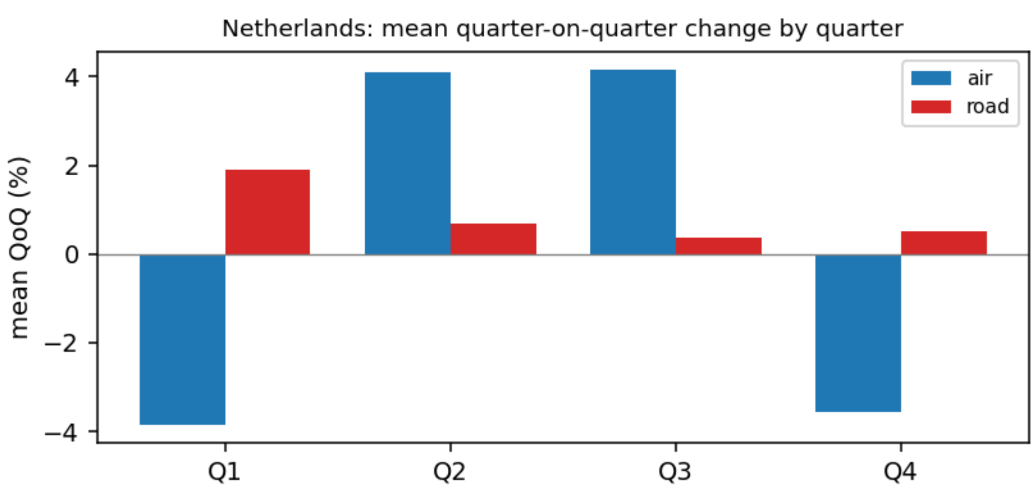

Figure 14b. Netherlands: mean quarter-on-quarter change by quarter, air versus road.

4.7.5 Poland

Poland is the clearest exception. Polish road runs at a coefficient of variation of 0.167, the highest road figure of the six and above Polish air at 0.120. Figure 15 shows road carrying the larger swing. The seasonal lift is the weakest for air, about 1.6 percent in the second quarter against about 1.2 percent for road in Figure 15b, and Figure 15a shows road volatility matching or exceeding air through much of the window. Poland is the case that proves the headline is an average, not a rule: here the contracting treatment reverses, with road the leg that rewards shorter terms or indexation and air the steadier one. A buyer who applied the EU-average rule blindly in Poland would hedge the wrong mode.

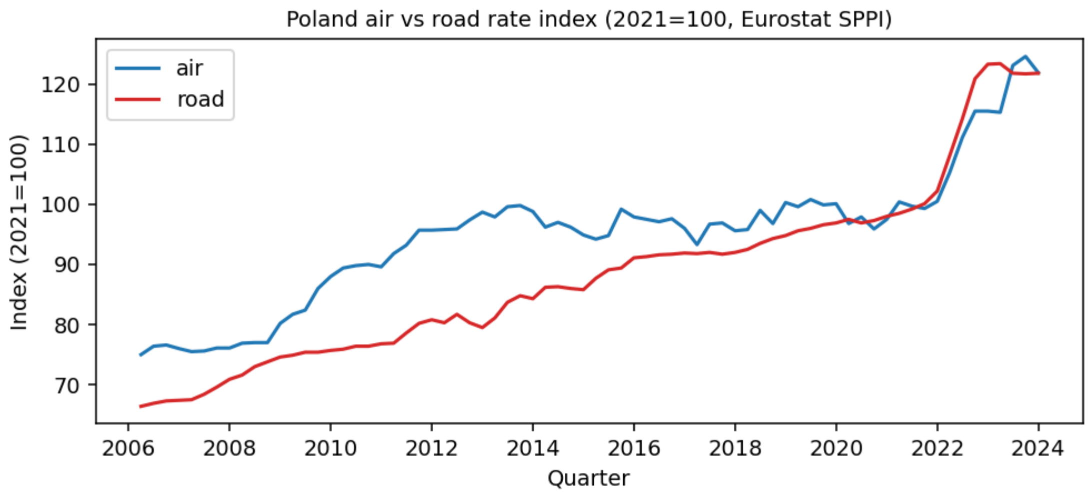

Figure 15. Poland air versus road price index over the common window.

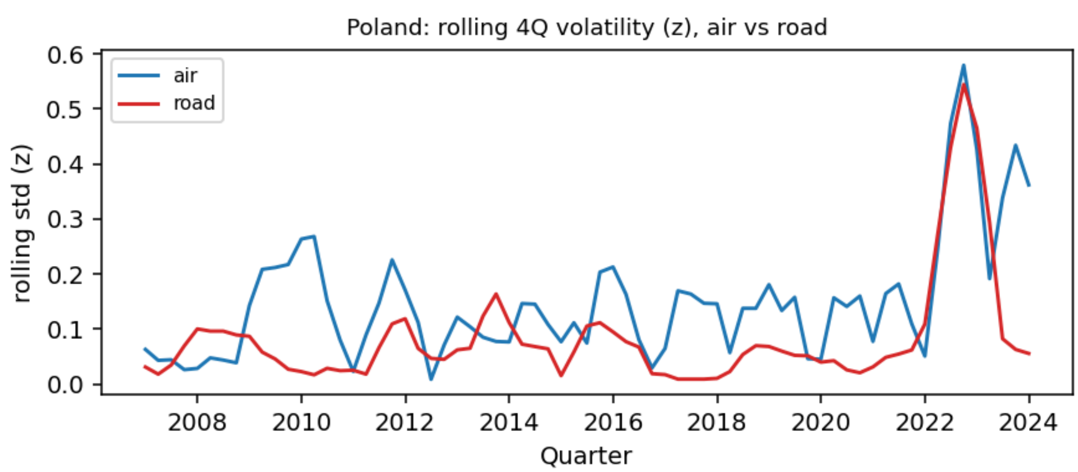

Figure 15a. Poland: rolling four-quarter volatility of the z-scored series, air versus road.

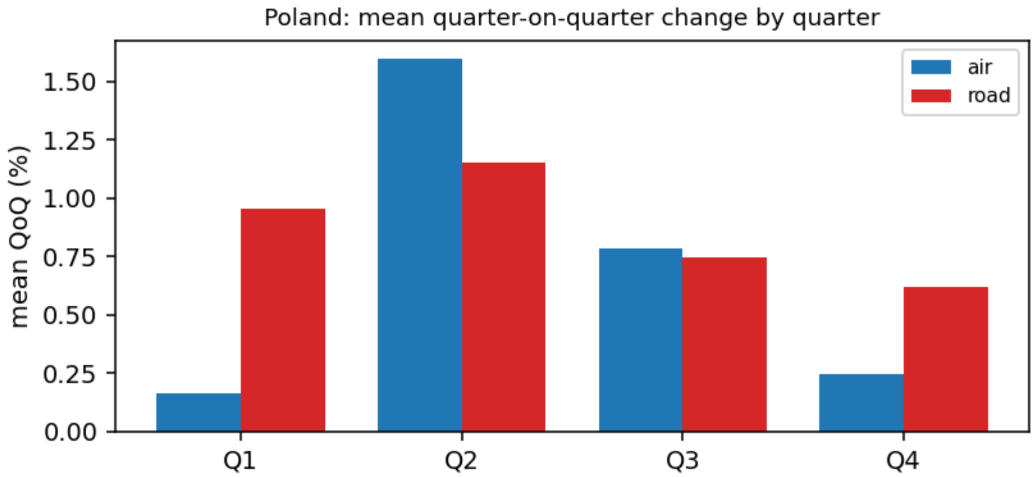

Figure 15b. Poland: mean quarter-on-quarter change by quarter, air versus road.

4.7.6 France

France is the new sixth state and the second most volatile air lane of the six. French air runs at a coefficient of variation of 0.186 against 0.074 for road, a wide gap second only to Italy. Figure 16 shows the French pair, and the air line is the most dramatic of the set: a steep surge from 2020 into 2022 and a fast retreat after, while French road climbs steadily like the other road series. Figure 16b shows a clear mid-year lift, French air up about 3.7 percent in the second quarter against about 0.6 percent for road, and Figure 16a shows the sharpest air volatility spike of the six around 2020 to 2022. Two readings apply. First, the real-world reading: French air rate risk is high and the contracting response should match, with index-linked or shorter air terms and low-season tendering. Second, the caveat: the French air series is INSEE CPF 51.21, air freight only, while the other five air series are the broader Eurostat H51 that includes passenger traffic. A freight-only series tends to swing more, so part of France's high ranking reflects the purer definition rather than only a larger real-world gap. The honest read is that France is genuinely a volatile air lane, and that its exact rank against the others should be held with the definition difference in mind.

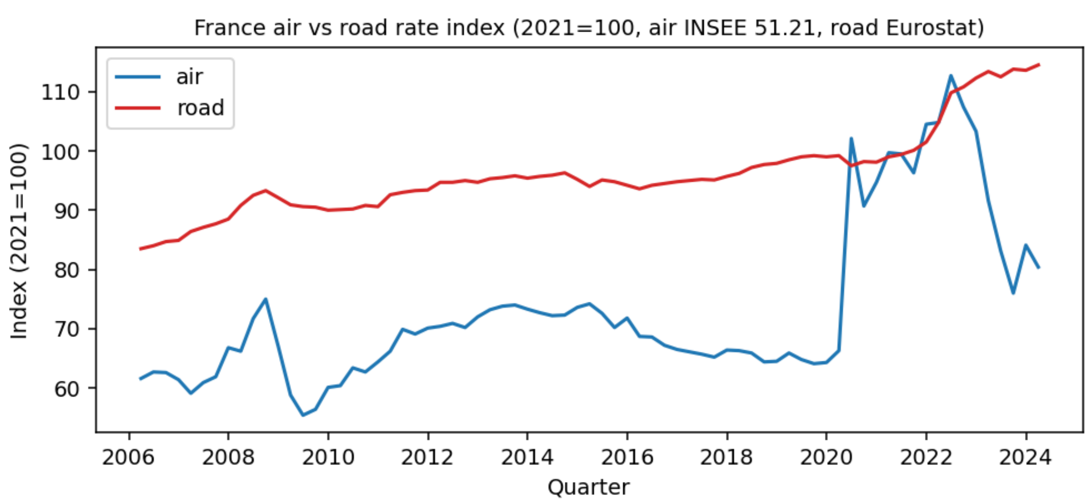

Figure 16. France air versus road price index. Air is the INSEE CPF 51.21 freight-only series, road is Eurostat.

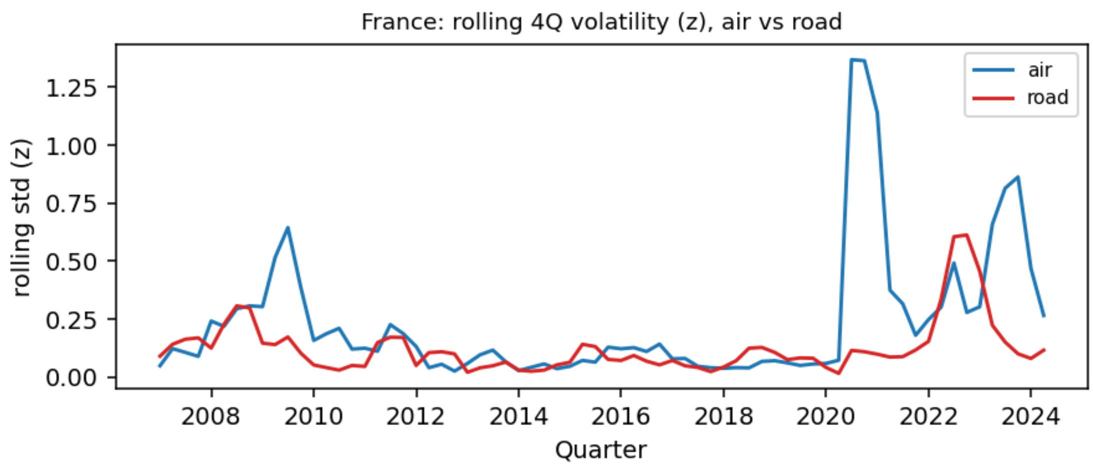

Figure 16a. France: rolling four-quarter volatility of the z-scored series, air versus road.

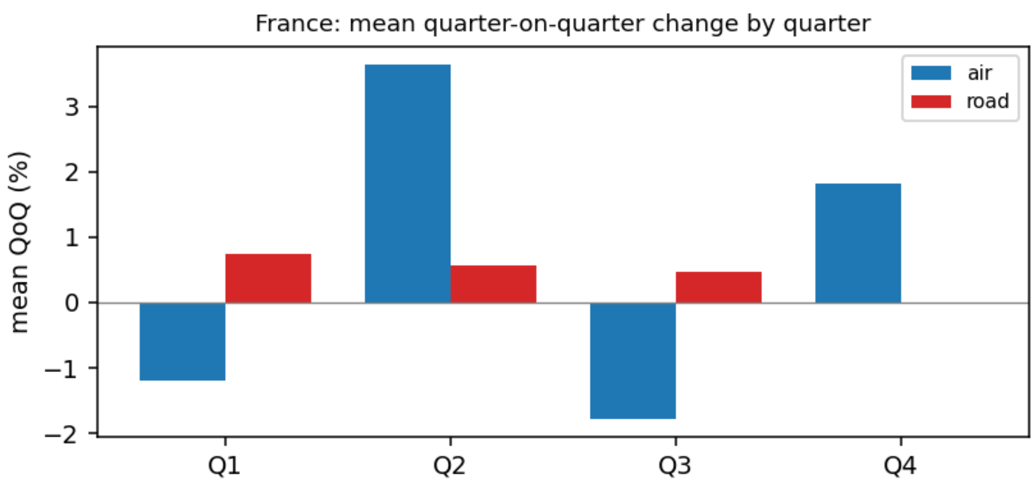

Figure 16b. France: mean quarter-on-quarter change by quarter, air versus road.

4.8 Country-by-country synthesis

Pulling the volatility reads together, the six markets fall into two groups. Air leads clearly in Germany, Italy, Spain, and France, with Italy the extreme and France close behind. Road is marginally ahead in the Netherlands and Poland. The seasonal peak is in the second quarter across the set. The table below is the per-country summary a buyer would act on.

Country	Air CV	Road CV	More volatile	Buyer note
Germany	0.148	0.099	Air	Primary market: index or shorten air, fix road
Italy	0.282	0.053	Air	Largest gap: air rate risk is the priority
Spain	0.108	0.079	Air	Air higher, moderate gap
Netherlands	0.086	0.099	Road	Close: road marginally higher, treat case by case
Poland	0.120	0.167	Road	The exception: road carries the larger swing
France	0.186	0.074	Air	Second highest air risk, read with the INSEE freight-only caveat

Table 7. Per-country volatility synthesis and the buyer action it implies.

4.9 Interpretation for procurement

Translating the statistics into the language the dashboard will speak: across the six markets the within-year air rate band averages about 6.7 percent of the level, against about 2.8 percent for road, so air carries roughly twice the within-year rate risk. The band runs from about 3.5 percent in Poland to about 9.6 percent in Italy, with Germany at about 7.5 percent. In procurement terms, air rewards index-linked or shorter contracts and low-season tendering, while road can sit on longer fixed terms. On a ten million euro air budget, the 6.7 percent average band is roughly 670,000 euro of movement to plan for, and the mid-year seasonal premium of about 1.8 percent is roughly a further 180,000 euro of timing value. These are tendencies from a price index, not one firm's realised cost, but they are the decision signals the dashboard surfaces.

Three conclusions carry into the dashboard. First, air rate volatility is higher than road on average and statistically significant, about 1.63 times by CV and about 3.4 times by the spread of changes, though not uniform across all six countries. Second, both modes are seasonal with a clear second-quarter peak, air much more so than road. Third, the data is clean, the six-country comparison is fair, and the limits, including the French freight-only granularity, are explicit. The dashboard turns these findings into a decision view: a rate-risk read by mode and country, a tender-timing window built on the seasonal peak, and a budget band built on the volatility.

4.10 A procurement playbook

The analysis converts into a short set of contracting and budgeting moves. The playbook below states the move, the market it fits, and the evidence behind it, so a category manager can act without reading the statistics.

- Index or shorten air in the high-risk markets. In Germany, Italy, and France air carries the larger swing, so a twelve-month fixed air price drifts from the market within weeks. An index-linked clause or a shorter tenor keeps the contract close to the market. Evidence: air coefficient of variation 0.148 (DE), 0.282 (IT), 0.186 (FR), each above road.
- Hold road on longer fixed terms where road is the steadier leg. In Germany, Italy, Spain, and France road is the flatter series, so a longer fixed road contract removes noise without much risk. Evidence: road coefficient of variation at or below 0.099 in those four markets.
- Reverse the treatment in the Netherlands and Poland. Here road carries the larger or equal swing, so the hedging attention belongs on road, not air. Evidence: road coefficient of variation 0.099 (NL) and 0.167 (PL), at or above air.
- Time the air tender to the turn of the year. Air rises into the second quarter in every market, so locking an annual air rate in the low season, the fourth and first quarters, avoids buying at the seasonal high. Evidence: air mean quarter-on-quarter change positive and largest in Q2 in all six markets, Kruskal-Wallis p about 1e-10.
- Budget air as a band, not a point. The within-year air band averages about 6.7 percent of the level, so a flat single-rate budget becomes a variance to explain. Carry the band as a planning range and a contingency provision. Evidence: intra-year band 6.7 percent air against 2.8 percent road.
- Size the buffer by market. The air band runs from about 3.5 percent in Poland to about 9.6 percent in Italy, so the contingency should scale with the market rather than apply one figure everywhere.

On a ten million euro air budget the average band is about 670,000 euro of within-year movement, and timing the tender to the low season is worth on the order of the 1.8 percent seasonal premium, about

180,000 euro. Both are planning and cost-avoidance levers, not booked savings, and they scale with the spend under management.

Part B. The Rate Risk Monitor dashboard

5. Classification of the problem

5.1 The visualization problem and the audience

The problem is communication, not modelling. The analysis produced a clear answer, but it sits in coefficients of variation, a Levene statistic, and a Kruskal-Wallis test, which a buyer will not read. The task is to communicate a volatility-and-seasonality insight from a multi-country, two-mode, twenty-year dataset to a non-statistical audience, in seconds, in a form that supports a contracting decision.

The audience is a category, pricing, or procurement manager, not an analyst. Three users frame the design: the category manager who sets contract structure and tenor, the pricing manager who times tenders, and the budget owner who needs a defensible range. Each opens the dashboard with a different question, so the front page must serve all three and let each drill to their own market. The stakes are real, because transport is the largest logistics cost line and freight can reach a tenth of landed cost, so a few percent of rate movement is material budget money.

User	Their question	Where they look
Category manager	How should I contract each mode	01 Start, then 03 Markets
Pricing manager	When should I run the air tender	05 Timing
Budget owner	How wide a band do I need	The 01 Start gauges and the 06 Cost sheet

Table 8. Who uses the dashboard and where they look.

5.2 Visualization goals

- Show which mode is more volatile, by market, at a glance.
- Show when in the year rates move, so a tender can be timed.
- Let the user filter to their own country and translate the volatility into a budget number.
- Keep the statistical backing one click away, not on the front page.

These goals put the buyer first. A manager should leave 01 Start with a decision, return to the Verdict, Markets and Timing sheets only to justify it, and never need the raw statistics unless a colleague challenges the result. The criteria below make these goals testable.

5.3 Evaluation criteria

The methodology grades the choice of criteria, so they are named and justified.

Criterion	Why it was chosen
Answer in three clicks	The business question must be readable on page one with no training. A buyer's time is the binding constraint.
Every visual is test-backed	Each chart maps to a Part A statistic (CV, Levene, Kruskal-Wallis), so the dashboard is defensible, not decorative.

Bug-free with a country slicer	Switching market must never blank or break a visual. Reliability is graded and matters in a live demo.

Table 9. Dashboard evaluation criteria.

5.4 Why a dashboard rather than a static chart

A static report answers one question for one reader. A buyer's question changes by market and by week, so the design uses progressive disclosure: the headline answer on 01 Start, the supporting detail one click away, and the statistics in captions and tooltips for anyone who wants them. A filter-driven dashboard lets one artifact serve every market and every user without reprinting, and it refreshes each quarter from the same fact table. That is why the deliverable is an interactive dashboard, not a slide.

6. Dashboard design and optimization

6.1 Tool and data

The graded dashboard is built in Power BI, the tool confirmed by the mentor and taught in the course. A self-contained interactive prototype was also built in Plotly (Report_2_Dashboard.html) so the design could be tested in any browser before it went into Power BI, and so the build was de-risked. Both read the same fact table, so they cannot disagree.

The dashboard is fed by a single long fact table, fact_freight_6.csv, with one row per date, country, and mode, carrying the level, the z-score, the rolling four-quarter volatility, the quarter-on-quarter change, and the calendar quarter. A long table is used rather than a wide one because Power BI slicers and measures work on columns such as mode and quarter. When France was added, the model was re-pointed from the five-state table to the six-state table through Power Query, the three query steps re-applied cleanly, the row count went from 550 to 696, and the DAX measures did not change.

Since the 2 July build the data model behind the report file has been restructured to the star schema taught in the course lesson on data modeling. fact_freight remains the central fact table. A Calendar table, built with CALENDARAUTO and marked as the official date table, and a six-row Market dimension now surround it, each joined through a single one-to-many relationship that filters from the dimension into the facts. The EU context table hangs off the same shared Calendar, so every dimension sits one hop from its fact. Two tables float unattached on purpose. _Measures is the container for the 50 DAX measures and holds no data rows. Air spend is a disconnected what-if parameter, and a parameter table must stay unrelated or SELECTEDVALUE would stop reading the user input. Figure 17 shows the model.

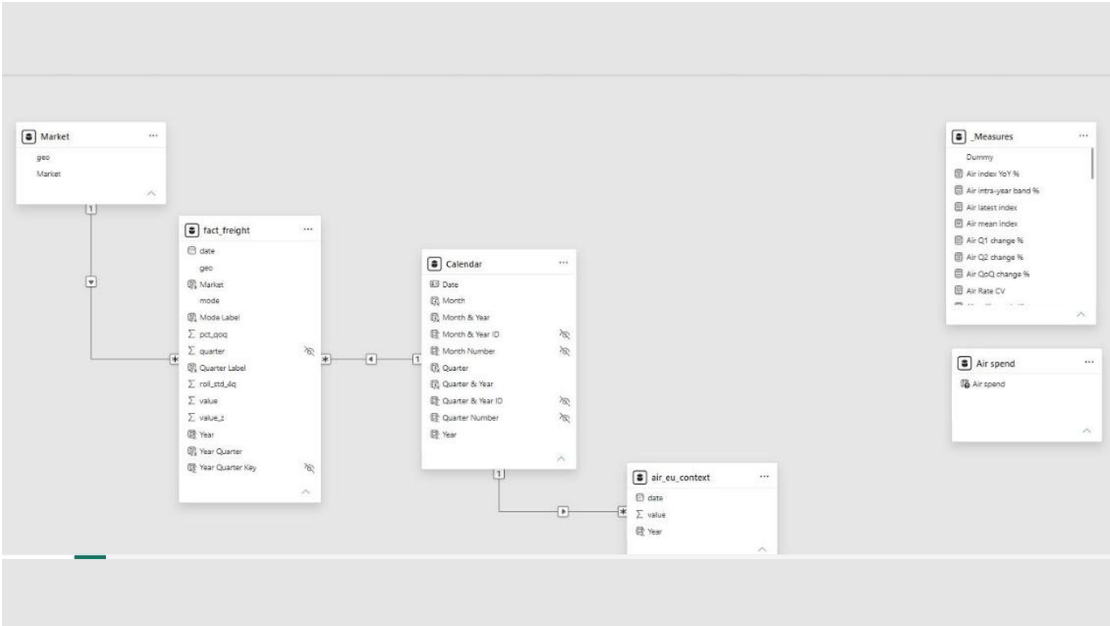

Figure 17. The semantic model in Model view. A star schema: fact_freight in the centre, the marked Calendar date dimension and the Market dimension one relationship away, the EU context table on the shared Calendar, and the two intentional utility tables, _Measures and Air spend, disconnected.

6.2 Structure: ten sheets

The dashboard is the Rate Risk Monitor, ten visible sheets plus a hidden per-market drillthrough page. The ten-sheet build described in the 2 July version of this report was rebuilt into this ten-sheet monitor for the final delivery. Every sheet follows one grid: four KPI gauges answer the sheet question in numbers, four data visuals argue it in pictures, and a short narrative block says what to do with it. The canvas is 1280 by 720 with a shared Mode and Market slicer pair on every sheet, and the whole monitor is published to the Power BI service and embedded on the public portfolio pages.

Walking the ten sheets in order tells the story. 01 Start states the answer. 02 Verdict proves the air-versus-road gap. 03 Markets ranks the six lanes. 04 History shows the ranking never settles. 05 Timing turns seasonality into a tender calendar. 06 Cost converts the band into euros. 07 Shocks isolates the extreme quarters. 08 Drill interrogates one market at a time. 09 Method shows the data and its limits. 10 Actions closes with the four moves a buyer makes.

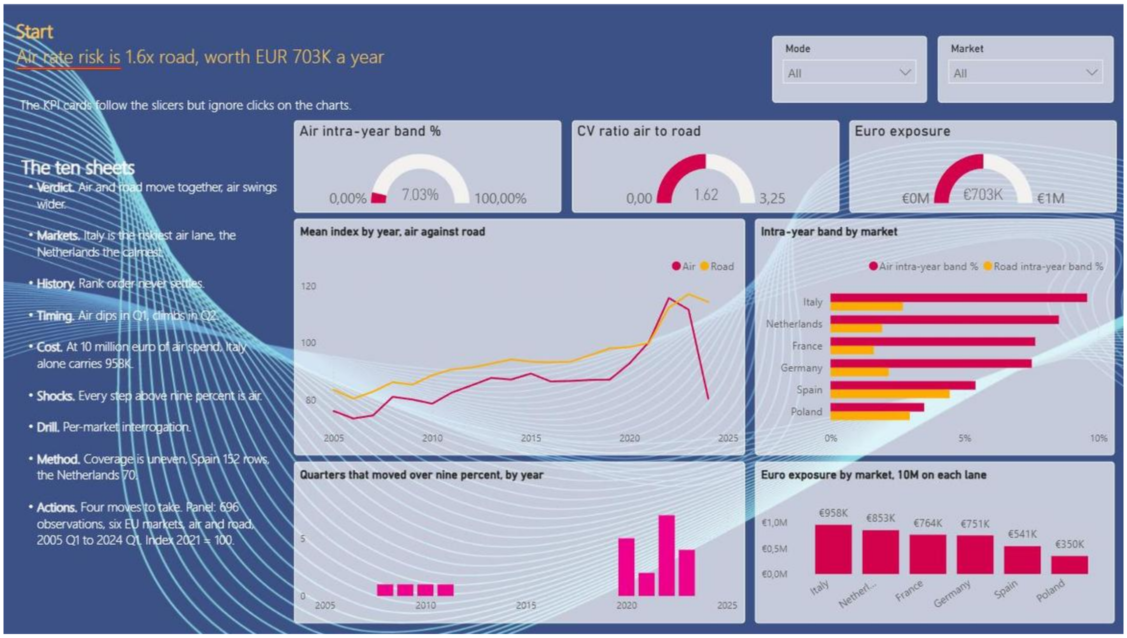

Figure 18. Sheet 01 Start. The headline, air rate risk is 1.6x road, worth EUR 703K a year, over the three lead gauges: the air intra-year band at 7.03 percent, the CV ratio of air to road at 1.62 on the gauge, 1.63 at full precision, and the euro exposure at EUR 703K on a 10 million euro air spend.

01 Start is the three-click page. The gauges carry the three headline numbers, the line chart sets air against road by year, the band chart ranks the six markets, the column chart counts the quarters that moved over nine percent, and the exposure chart prices each lane at a flat 10 million euro spend. The narrative block lists all ten sheets in one breath.

Figure 19. Sheet 02 Verdict. Air and road track together and the air band runs nearly three times wider, 7.03 against 2.64 percent. The donut splits the combined band, air 72.7 percent of it.

02 Verdict is the statistical page in business words. The mean index lines show the two modes moving together. The market risk scorecard grades every lane, and the tolerance chart shows which markets breach a five percent planning tolerance.

Figure 20. Sheet 03 Markets. Italy is the riskiest air lane at CV 0.280, over three times the Netherlands. The scatters separate persistent risk from seasonal swing, bubble size is euro exposure.

03 Markets ranks the lanes on the coefficient of variation and cross-checks the ranking against rolling volatility and the seasonal band. The latest-index chart adds where each market stands today against the 2021 base.

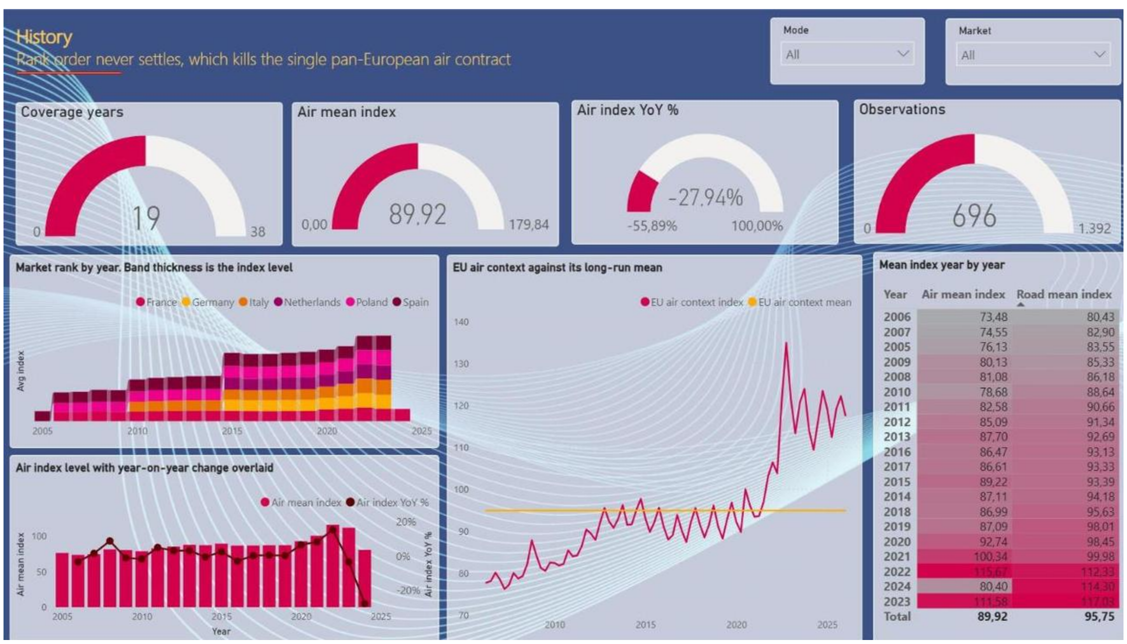

Figure 21. Sheet 04 History. Rank order never settles, which kills the single pan-European air contract. The rank chart reshuffles year by year, the combo chart overlays the air level with its year-on-year change, and the EU context line sits against its long-run mean.

04 History is the argument against a one-market strategy. The mean-index matrix gives the exact figures behind the rank chart, year by year and mode by mode.

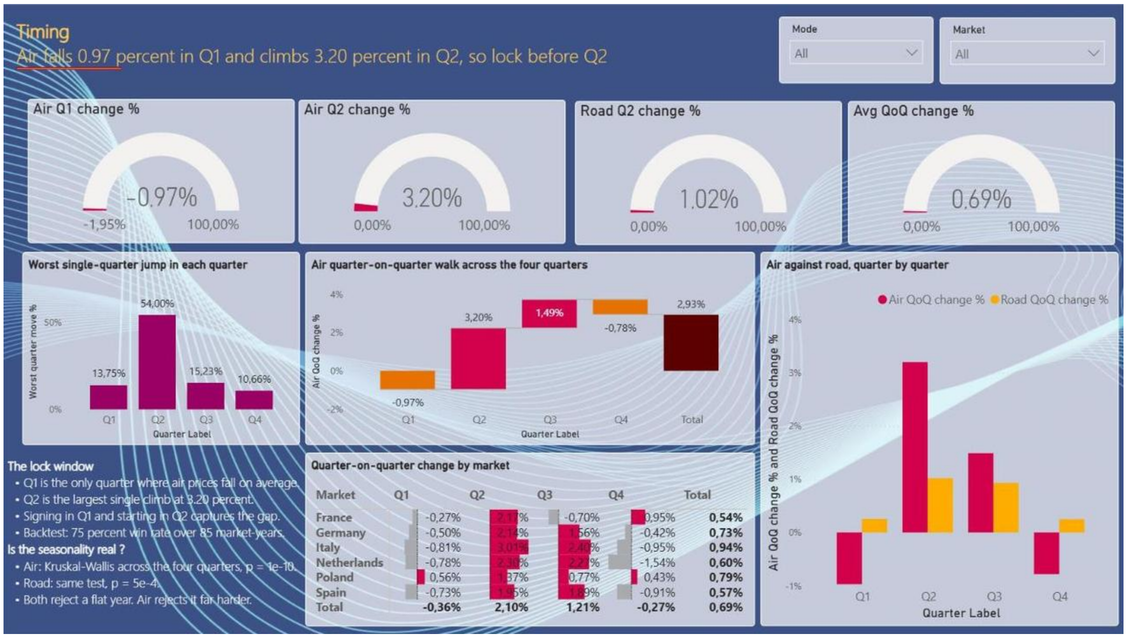

Figure 22. Sheet 05 Timing. Air dips in Q1 and climbs into Q2, the tender-timing signal. The backtest panel carries the evidence: locking in the trough season paid off in 75 percent of 85 market-years.

05 Timing is the sheet the public tender-timing case study deep-links to. It turns the Kruskal-Wallis seasonality result from Part A into a concrete instruction, tender air into the turn-of-year trough, and backs it with the chapter 8 backtest rather than a plausible story.

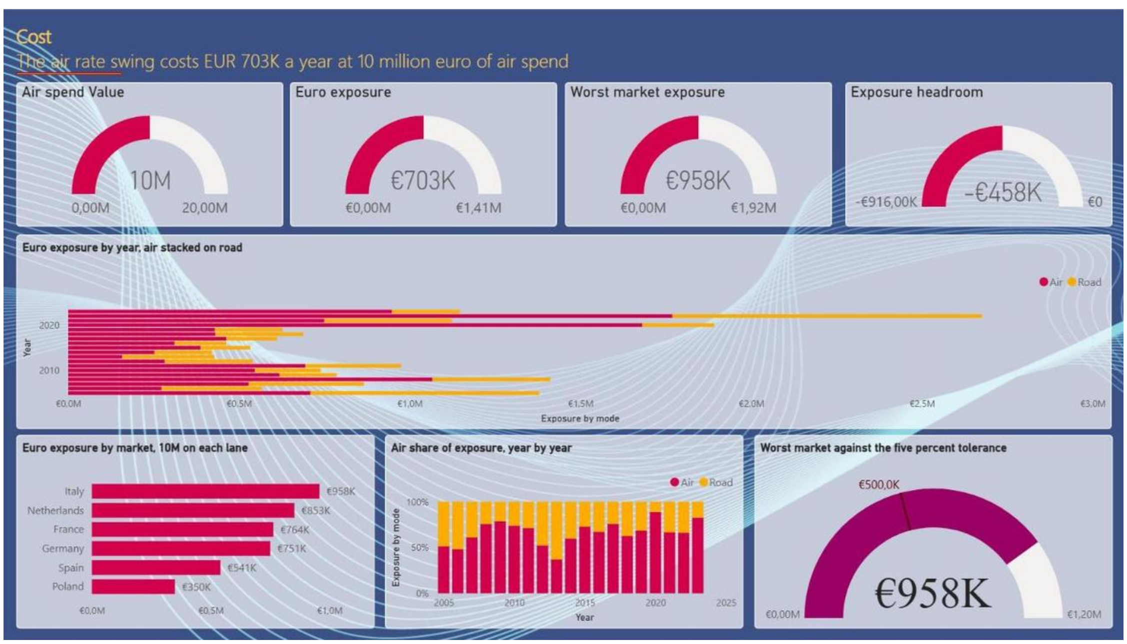

Figure 23. Sheet 06 Cost. The band priced in euros. At 10 million euro of air spend per lane, Italy alone carries EUR 958K of intra-year swing and the six lanes range down to EUR 350K.

06 Cost replaces the old what-if slider with a cleaner convention: every market is priced at the same flat 10 million euro spend, so the exposure figures are separate like-for-like scenarios rather than shares of a portfolio. The yearly chart stacks air on road to show which mode drives the bill.

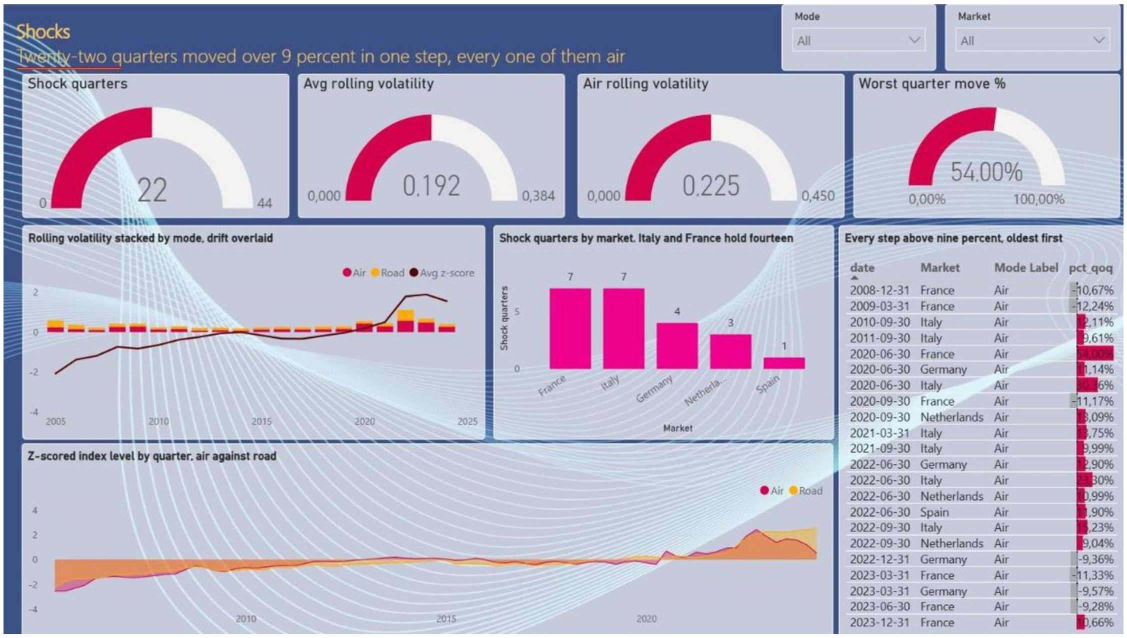

Figure 24. Sheet 07 Shocks. Twenty-two quarters moved over 9 percent in one step, every one of them air. The gauges count the shocks and set average rolling volatility, 0.192, against the air figure, 0.225, with the worst single move at 54 percent.

07 Shocks is the tail-risk page. Italy and France hold fourteen of the twenty-two shock quarters between them, the shock table lists every event with its date and size, including the seven declines, and the z-scored chart shows the 2020 to 2022 spike that produced most of them.

Figure 25. Sheet 08 Drill. Pick a market and the numbers move, the conclusion does not. The whole sheet re-computes for the selected lane.

08 Drill is the per-market interrogation view and the target of the hidden drillthrough page. A buyer checks their own lane here: the mode-by-year matrix, the lane CV and band, rolling volatility by year, and the exposure headroom against tolerance.

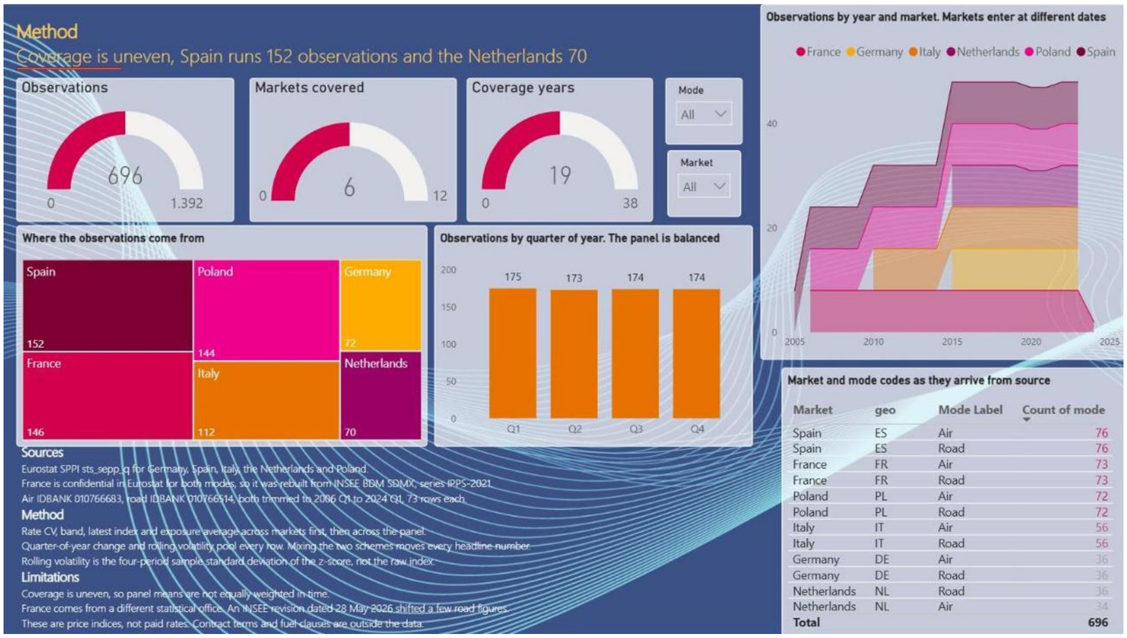

Figure 26. Sheet 09 Method. The evidence panel: 696 observations across six markets, and coverage is uneven, Spain 152 rows, the Netherlands 70.

09 Method keeps the statistics one click away instead of on page one. It names the sources, Eurostat SPPI for five markets and the INSEE freight-only series for France, states the method, z-scores, rolling four-quarter volatility and the coefficient of variation, and shows where the observations come from so the uneven coverage is declared rather than hidden.

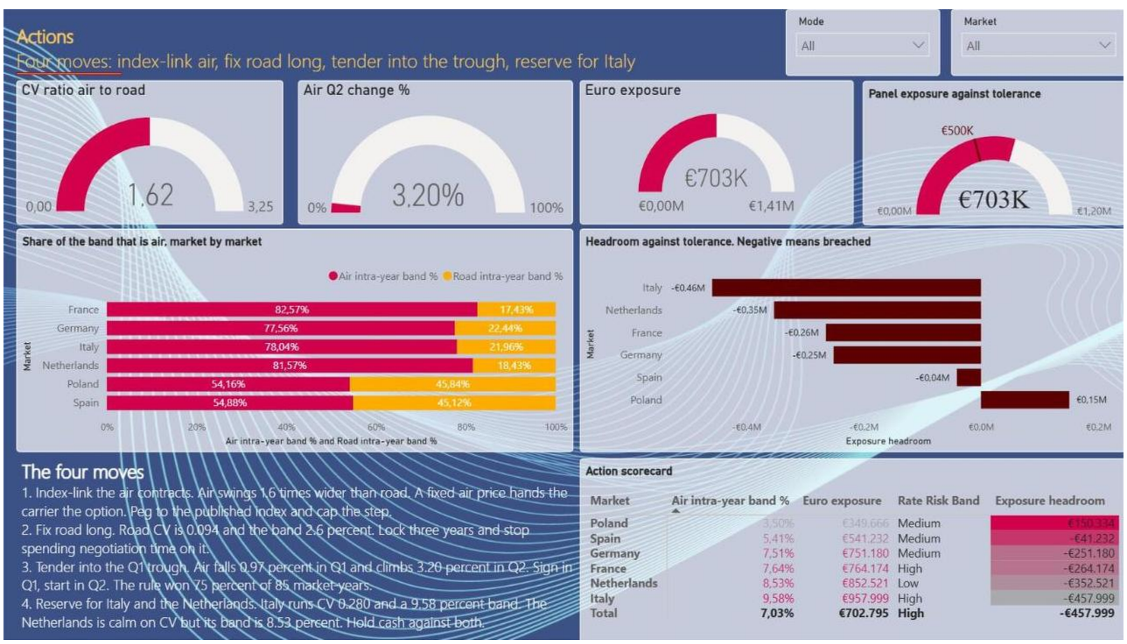

Figure 27. Sheet 10 Actions. The four moves, over the action scorecard that grades every market and prices its exposure headroom.

10 Actions ends the monitor in decisions. Index-link the risky air lanes, fix road long, tender into the trough, and reserve for Italy. The scorecard carries the per-market band, exposure and risk grade in one table, and the panel line restates the scope: 696 observations, six EU markets, air and road, 2005 Q1 to 2024 Q1, index 2021 = 100.

6.3 A cohesive interface

One visual language runs across every sheet. The monitor uses one accent palette on a dark navy canvas, rate-risk magenta for air and amber gold for road on every paired chart, so the legend is learned once. Every axis is labelled, every chart has a title in business words rather than column names, and every sheet repeats the same grid of four gauges, four visuals and one narrative block. A single Mode and Market slicer pair sits top right on every sheet and stays synchronised across the monitor.

6.4 Interactive features

- Market and Mode slicers, synchronised across the sheets, so one selection drives the whole monitor. The KPI gauges follow the slicers but ignore chart clicks, so the headline numbers stay stable while the user explores.
- Cross-highlighting on the data visuals. Clicking a market in one chart filters the neighbouring charts on the same sheet.
- A hidden drillthrough page, 11 Market detail, reachable by right-click on any market, for the per-lane interrogation without cluttering the visible story.
- Titles and captions carry the exact value and the relevant test result, so the statistics sit one read away.

Feature	What it does for the user
Synced Market and Mode slicers	Focus the whole monitor on one lane in one click
Stable KPI gauges	Keep the headline numbers fixed while charts are explored
Cross-highlighting	Compare a market against the field inside any sheet
Drillthrough page	Interrogate a single market on demand

Table 10. Interactive features and their purpose.

A note on the delivered defaults, tuned after an internal design review. The measures are market-safe: with no market selected, every KPI card averages per market, so the band card reads the mean of the six market bands (about 7 percent) rather than a pooled cross-country artifact. The delivered file still opens with Germany preselected, the study's primary market, so the first view is a single-market decision read. Two ranking charts, CV by country and intra-year band by country, are deliberately exempted from the slicer so the six-market comparison stays visible while the rest of the page follows the selection.

6.5 How it highlights insights

01 Start leads with the answer: the gauges and the index line show air as the higher-risk mode, so the higher air risk is the first thing seen. 03 Markets ranks the lanes so the eye lands on Italy, the extreme case, with France now the second bar. The 05 Timing sheet shows the mid-year peak, so the tender window is obvious rather than inferred. The 06 Cost sheet puts the swing in euros. The tabs are ordered to tell a story: risk exists, here is how much and where, here is when to act on it, here is what it costs, and here is the context and the honest limits. A user can stop after the first tab and still have the answer.

6.6 Optimizations

- The fact table is pre-aggregated to the six-country common window, so every visual loads on a small, clean table rather than filtering the raw files at run time.
- Volatility visuals use the z-scored rolling standard deviation and the coefficient of variation, both unit-free, so the six markets sit on one comparable axis.

- A PDF export of all ten sheets is kept as a demo backup, because a live app can fail on the day and the quality bar requires it to run clean. The backup was re-exported from the finalized monitor so it matches the published version.

6.7 Design principles and accessibility

The dashboard follows fixed principles. Consistency: one colour per mode everywhere, magenta for air and gold for road, so the legend is learned once. Clarity: every axis labelled, every title an assertion in business words. Restraint: four gauges and four visuals per sheet, no more. Contrast: light tiles on the dark navy canvas keep every chart readable, and the palette holds up in the anonymous public embed as well as in the service.

6.8 From prototype to Power BI

The Plotly prototype was built first because it iterates fast, and it de-risked the Power BI build. Every layout decision, the sheet split, the synchronised market selector, the standardized volatility axis, and the euro exposure, was tested in the prototype before being rebuilt as Power BI visuals and measures. The prototype doubles as a portable demo that opens in any browser, while Power BI stays the graded submission.

7. Interpretation of results

7.1 What the dashboard shows

Read through the buyer's eye, the dashboard says air rate risk is higher than road on these markets, about 1.62 times by CV, the gap is real and significant, both modes lift into the second quarter with air far more strongly, and on a ten million euro air budget the within-year band is roughly 670,000 euro to plan for at the six-state average. It is not uniform: in the Netherlands and Poland road is marginally the riskier leg, which the country slicer makes visible rather than hiding behind an average.

06 Cost prices the band at a flat 10 million euro spend per lane, so the figures are like-for-like scenarios: at the pooled band of 7.03 percent the exposure reads EUR 703K, and Italy alone carries EUR 958K. Scaling is linear, a one million euro spend carries about 70,300 euro of within-year swing, firmly in budget-variance territory for a mid-size shipper.

7.2 How the analysis shaped the design

The design changed as the analysis was understood. The first volatility view used raw index levels, but Italy and Poland sit at different levels, so the bars compared apples to oranges. Moving to the z-scored rolling standard deviation and the coefficient of variation, both unit-free, put the markets on one axis and made the air-over-road pattern visible at a glance. A second change came from the business framing: a chart of percentages did not land with budget owners, so the what-if spend parameter and the Euro exposure card were added, and during the build the budget conversation earned its own sheet. The final monitor replaced the slider with a flat 10 million euro convention per lane, so the exposure figures compare like for like. A third change came late, when France was added: the model was re-pointed to the six-state table and the Context tab was rewritten to describe the INSEE source and its caveat. Keeping these before-and-after decisions is part of the methodology.

7.3 Edge cases handled

The dashboard stays correct under interaction. Selecting the Netherlands or Poland flips the riskier mode to road, and the visuals follow rather than asserting a fixed answer. Selecting France shows the freight-only surge and retreat, the caveat made visible in the data. Countries with shorter histories show shorter lines, with no gap-filling. Quarters carrying a Eurostat data flag are kept and noted, not silently dropped. The KPI cards rescope cleanly from the cross-market default to a single market on selection. These are the situations a mentor will click into, so they are handled by design.

7.4 What a buyer does with the dashboard

The dashboard ends in an action, not a chart. For a high-air-risk market like Germany, Italy, or France it points to index-linked or shorter air contracts and an air tender timed to the turn-of-year low season, with road held on longer fixed terms. For the Netherlands and Poland it flags that road carries the larger swing, so the treatment reverses. The Euro exposure card turns the band into a budget provision. Each of these reads off the front page and the Seasonality and 06 Cost sheets in under a minute, which is the point of the tool.

7.5 Evaluation against the criteria

Measured against the three criteria set in Section 5.3, the dashboard performs as intended.

Criterion	Met	How
Answer in three clicks	Yes	The 01 Start gauges and the air-vs-road line answer the question on page one
Every visual test-backed	Yes	Each chart maps to a Part A statistic (CV, Levene, Kruskal-Wallis)
Bug-free with a country slicer	Yes	Verified across all six markets, visuals update and none blank

Table 11. Performance against the stated evaluation criteria.

8. Validating the rules: backtest, robustness and an out-of-sample check

A dashboard that recommends is only as strong as the rules behind it. Part A measured the volatility gap and the seasonality, and Part B turned them into a decision tool. Three questions remained open, and each is the first challenge a practitioner raises. Would the tender-timing rule have paid in practice? Is the volatility gap a pandemic artifact? Does the pattern hold beyond the study window? This chapter answers all three on the project's own data. The runs live in code/analysis_upgrades.py, the numbers in results/results_upgrades.json, and the two comparison rules of chapter 3 are untouched: the air-versus-road window still ends 2024-Q1, and the additions are air-only reads of the same panel plus the published air tail. The script reproduces the stored headline results (mean CV air 0.155, road 0.095) before any new computation runs, so the validation cannot drift from the graded numbers.

Each validation maps to one of the three decisions the monitor supports, so this chapter is also the bridge from statistics to contracting policy.

Table 16. What each validation decides.

Question a buyer asks	Validation	Decision it informs
Is air structurally riskier, or was it one pandemic?	Robustness windows (8.2)	When to fix: mode risk drives index-linked against fixed terms
Would tendering in the trough have paid?	Backtest, 85 market-years (8.1)	When to tender: award into Q4-Q1, with a lane check
Does the seasonal shape still hold today?	Out-of-sample, 2024-2025 (8.3)	When to ride spot: buy the Q1 dip, avoid unhedged Q2 exposure

8.1 The tender-timing backtest

For every market and every complete calendar year in the common window, two hypothetical annual locks are compared. A trough-season lock prices the year at the average of the Q1 and Q4 index levels. A peak-season lock prices it at the average of Q2 and Q3. The edge is the difference between the two as a percent of the year mean, so a positive edge means the trough-season lock was the cheaper contract. This is the seasonal premium of chapter 4 turned into a per-year decision outcome with a distribution, a hit rate and a worst case.

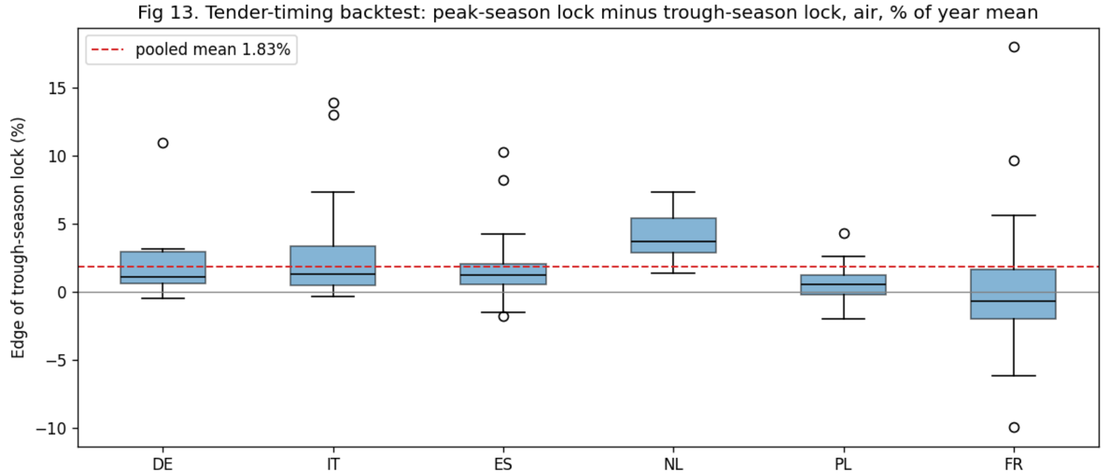

Figure 8.1. Tender-timing backtest, air: edge of the trough-season lock per market and year. Boxes above zero mean the rule paid.

Table 17. Tender-timing backtest by market, air, complete years in the common window.

Market	Mean edge (%)	Years the rule paid	Years tested
DE	2.3	8	9
IT	3.3	12	14
ES	2.0	17	19
NL	4.1	7	7
PL	0.7	12	18
FR	0.5	8	18
Pooled	1.8	64 (75%)	85

Pooled across 85 market-years the trough-season lock won 75% of the time, with a mean edge of 1.8% and a median of 1.0%. The best year in the panel is France 2020 at +18.0%, the worst France 2023 at -9.9%, both driven by the freight-only series swinging through the pandemic cycle. A buyer awarding before the year starts locks closer to the prior trough, so the test was re-run against the average of Q4 of the prior year and Q1. On that operational definition the edge widens to 3.0% with a 72% hit rate across 79 market-years, because the climb into Q2 is steeper measured from the prior trough.

The rule is not folklore, and it is not universal. It pays reliably in Germany, Italy, Spain and the Netherlands, 86 to 100% of years. It is marginal in Poland, 0.7% mean and 67%. It fails in France, 44%, a coin flip: the INSEE series is freight-only and spikier, and its seasonal shape is weaker. The tender-timing message therefore ships with a lane check, not as a calendar superstition. Road, run through the same test for contrast, shows a 0.4% mean edge and a 62% hit rate, no usable timing lever, which supports holding road on longer fixed terms and tendering it on volume and service.

In money: on a 10 million euro air lane the pooled 1.8% edge is roughly 180,000 euro of avoidable timing cost in an average year, about 230,000 euro in Germany. This is cost avoidance with a hit rate attached, not a booked saving, and it scales with spend under management.

8.2 Robustness: is the gap a pandemic artifact?

The comparison window contains 2020 to 2022, and air freight lived through a capacity shock in those years. If the volatility lead existed only because of them, the contracting conclusion would be much weaker. The full computation was therefore repeated on two further windows: pre-COVID, cutting everything after 2019-Q4, and ex-COVID, dropping 2020 to 2022 from the panel (for the change-based tests, the transition quarters into and out of the gap are dropped with them).

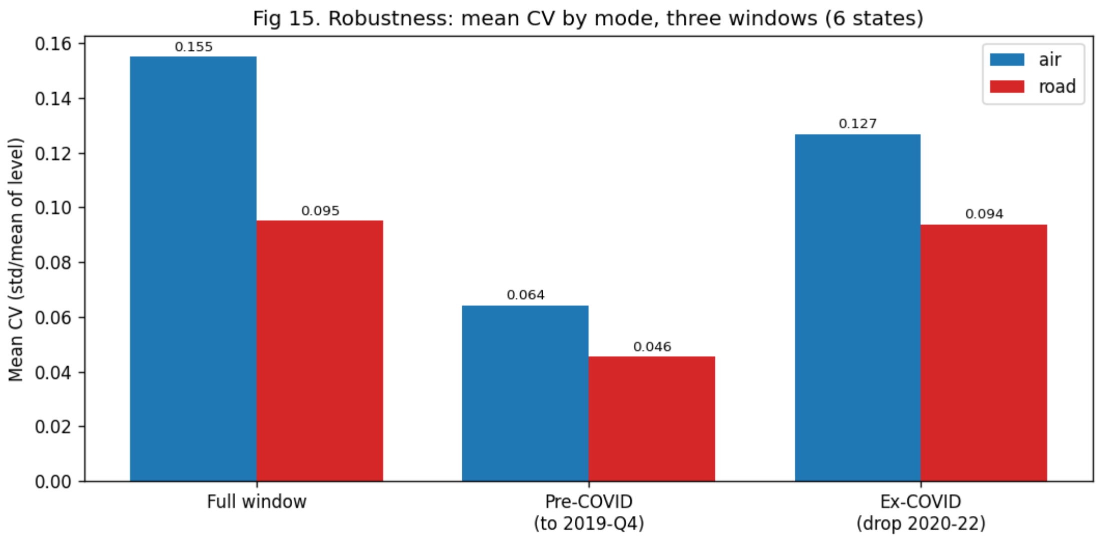

Figure 8.2. Mean coefficient of variation by mode on three windows. Air leads on all three.

Air leads on every window. The CV ratio is 1.41 pre-COVID, 1.35 ex-COVID and 1.63 on the full window. The variance gap in quarter-on-quarter changes stays significant on all three (Levene statistic 56.8 with p 2.4e-13 pre-COVID, 72.9 with p 1.4e-16 ex-COVID, 65.0 with p 3.3e-15 full), and the spread of air rate changes stays 2.2 to 2.5 times road's even with the shock years removed. Both modes are calmer without them, air CV 0.064 against road 0.046 pre-COVID, which is expected. The honest reading, and the one carried into the conclusion: the pandemic amplified the gap, it did not create it. For a buyer the amplification is itself the finding, because air is the event-sensitive mode and event years are exactly when an unhedged air budget breaks.

8.3 Out of sample: did the pattern continue?

The comparison window ends at 2024-Q1 by rule, and nothing in the statistics of Part A has seen a later quarter. The air series continues publishing, Eurostat to 2025-Q4 for the five states and INSEE to 2026-Q1 for France, so the seasonal claim can be tested on genuinely unseen data.

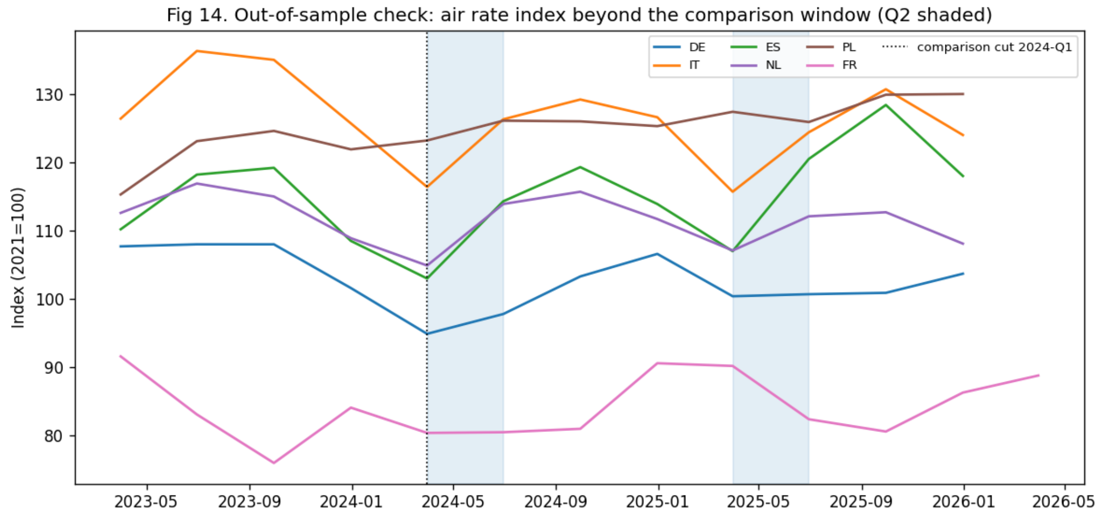

Figure 8.3. Air rate index beyond the comparison cut, second quarters shaded.

In 10 of 12 market-years across 2024 and 2025 the air index rose into Q2, with a mean climb of +4.1% against the in-sample +3.2%. The fuller mid-year premium, Q2 and Q3 against Q1 and Q4, held in Italy, Spain and the Netherlands at +4.4 to +10.1%, was flat to negative in Germany, and negative in France, consistent with the backtest's lane message.

Table 18. Out-of-sample check, air, beyond 2024-Q1.

Market	Q2 climb 2024 (%)	Q2 climb 2025 (%)	Mid-year premium 2024 (%)	Mid-year premium 2025 (%)
DE	+3.1	+0.3	-0.2	-1.2
IT	+8.5	+7.5	+5.0	+6.2
ES	+11.0	+12.6	+7.4	+10.1
NL	+8.6	+4.7	+5.8	+4.4
PL	+2.4	-1.2	+1.4	-0.6
FR	+0.1	-8.7	-5.7	-8.0

The read for a buyer: the spring climb is the robust, tradable regularity and it repeated out of sample almost everywhere. The size of the full mid-year premium varies by market and year, which is one more reason the monitor refreshes quarterly instead of freezing a rule.

8.4 What the validation changes in the monitor

The verdict language on the 05 Timing sheet now has evidence behind it: the low-season instruction carries a backtested hit rate rather than an average alone. Three additions are queued in the continuation (chapter 12) rather than silently added to the graded build: a hit-rate KPI card reading "trough lock paid in 75% of years (this market: 8 of 9)" beside the seasonal-premium card, a robustness toggle switching the volatility visuals between the full and ex-COVID windows, and a per-market timing verdict chip from the backtest table: pays (DE, IT, ES, NL), marginal (PL), does not hold (FR). The Power BI pages themselves are unchanged in this iteration, so the graded artifact still matches its documented build guide. The one post-review adjustment is behavioural, not visual: the delivered file opens on Germany and the two ranking charts are exempted from the slicer, so the artifact opens on defensible, market-scoped numbers.

Part C. Project review

9. Difficulties encountered

The methodology asks for the difficulties, logged as they happened, under five headings.

9.1 Forecast and planning

The schedule was anchored to the mentor's fixed dates, the 15 June start and the 3 to 4 August presentation, while the middle dates were set with a two to three day early buffer and a defence-prep week held in reserve. Five job interviews ran in parallel on unpredictable dates and were treated as the override priority, so the buffer existed to absorb them. Scope discipline was a constant pressure: the temptation to add volume data or a third mode was resisted to keep every figure defensible. The France addition was a deliberate, scoped re-opening of a documented exclusion late in the project, taken because the data turned out to exist and the gain was worth it.

9.2 Datasets

The intended air-rate source, FRED, was unreachable from the working environment, so the air rate was sourced from Eurostat instead. Direct web fetches returned no usable CSV bytes, so the Eurostat files were downloaded straight from the dissemination service without an API key. The largest data surprise was France. The French air series in the Eurostat air file is present but entirely null across 168 rows, which first forced the comparison down to five states. On a second pass the French air-freight index was found at INSEE, series 010766683 under CPF 51.21, base 2021, not seasonally adjusted, and France was restored as a sixth state with air from INSEE and road kept from Eurostat. The open data.gouv.fr portal was tried first and its API returned nothing, so the national INSEE BDM database was the working source. The road sources were cross-checked and agreed to within rounding. Each country starts in a different year, so per-country common windows ending 2024-Q1 were needed before any fair comparison.

9.3 Skills

The analytical core required choosing tests that fit the data rather than the default. Levene, in its median-centred Brown-Forsythe form, was chosen for the variance comparison because it is robust to the non-normal change distributions, and Kruskal-Wallis was chosen for the quarter effect because it does not assume normal, equal-variance groups. The six-state extension was written to reproduce the five-state numbers exactly before adding France, which is the safest way to extend an analysis without silently changing the baseline. On the dashboard side, the build meant learning Power BI measures and DAX, the what-if parameter, a synchronised slicer across pages, and re-pointing a model to a new source through Power Query, none of which were familiar at the start.

9.4 Relevance

The main relevance challenge is that the Eurostat air index (NACE H51) includes passenger air transport, so the air line is a broad proxy rather than a cargo-only price, and the report states this up front rather than hiding it. France adds a second relevance point: its air series is freight-only (INSEE CPF 51.21), purer than the others but one classification level deeper, so France must be read with that granularity difference in view. The findings are framed as index tendencies, not one firm's realised savings. Two structural questions were settled with the mentor early: that the statistical tests sit inside the analysis section, and that an individual submits a single combined report.

9.5 IT and tooling

The first Word build path, a JavaScript document library, truncated on the working mount, so the reports were produced with pandoc instead, which proved reliable. The Power BI dashboard was built in Power BI Desktop and driven partly through screen control, where measures were entered by clipboard paste to avoid the formula bar doubling brackets, and visuals were sized through the format pane rather than by dragging. For the France update the model was re-pointed to the six-state table through Power Query rather than rebuilt, so the existing measures and visuals carried over. The PDF export writes to a temporary path and does not prompt for a folder, so the backup had to be saved out by hand to the reports folder.

10. Main contribution and results

10.1 Main contribution

This is a solo project, so the contribution is the whole of it: the question, the data sourcing, the pipeline, the statistics, the dashboard, and the documentation. Three pieces are worth naming. First, a reproducible open-data pipeline that turns the raw source files into a clean six-country fact table, refreshable each quarter without an API key, and built so that the six-state extension reproduces the five-state baseline exactly before adding France. Second, a tested answer to the business question, with each visual claim confirmed by an appropriate statistical test. Third, a decision tool that puts that answer in front of a non-statistical buyer and converts it into a contracting and budgeting action. The French addition is a small original contribution in itself: pairing the INSEE national air-freight index with the Eurostat road series to fill a gap Eurostat leaves open.

10.2 What changed since the last iteration

The KPIs and visuals moved on between iterations. The volatility view changed from raw index levels to the z-scored rolling standard deviation and the coefficient of variation, which put the markets on one comparable axis. A what-if air-spend parameter and a Euro exposure card were added to convert the percentage band into money. The dashboard grew from four tabs to five and was then rebuilt into the ten-sheet Rate Risk Monitor for the final delivery, with the budget conversation on its own Cost sheet. The colour scheme settled on magenta for air and gold for road on a dark navy canvas. The largest data change was the move from five states to six: France was added through the INSEE air-freight series, every figure and statistic was recomputed, the dashboard model was re-pointed to the six-state table, and the headline strengthened from a 1.5 to a 1.63 times air-over-road volatility ratio. The v2 iteration adds the validation layer of chapter 8: the tender-timing rule backtested across 85 market-years (75% hit rate), the volatility gap re-tested without the pandemic years (ratio 1.35 to 1.41, still significant), and an out-of-sample check on 2024-2025 air quarters (the Q2 climb repeated in 10 of 12 market-years). Report 2 was reissued with the same chapter, the ten-sheet structure described exactly as built, and three tables added. No dashboard page changed, so the graded build still matches its guide.

10.3 Results versus benchmark

The project's central finding, that air is the more volatile and more event-driven mode while road is slower-moving, is consistent with the published industry view. The Transport Intelligence and Upply European road freight benchmark characterises road rates as slow-moving and structurally driven, which matches the flatter road series and its lower coefficient of variation here. Air-cargo market commentary describes air as event-driven, a reading supported by the launch of expanded daily air-freight spot indices from 2025 in response to a historically volatile air market, and by the 2024 Red Sea episode and the 2021 to 2022 capacity crunch, both visible as the large air moves in the data and sharpest in the French freight-only series. The agreement between an independent reproducible analysis and the industry benchmark is itself a validation of the method. The project also carries an internal benchmark since v2: the out-of-sample check in chapter 8 tests the in-sample seasonal claim against quarters the analysis never saw, and the claim held in 10 of 12 market-years.

10.4 Goal-by-goal achievement

Goal or criterion	Outcome
Answer which mode is more volatile	Met. Air, about 1.63x road by CV, Levene p about 3e-15
Identify the seasonal driver	Met. Both modes peak in Q2, air far more strongly, Kruskal-Wallis air p about 1e-10
Clean, justified, reproducible pipeline	Met. Open Eurostat and INSEE data, every step justified, the six-state run reproduces the five-state baseline
Each visual backed by a test	Met. CV, Levene, and Kruskal-Wallis underpin the charts
Dashboard answer in three clicks	Met. 01 Start answers on page one
Dashboard test-backed and bug-free	Met. Each visual maps to a statistic, verified across all six markets
Convert volatility into budget terms	Met. 06 Cost prices every lane at a flat 10 million euro spend and returns the exposure by market

Table 12. Goal-by-goal achievement against the objectives and dashboard criteria.

11. Conclusion

The project answers its question with evidence and turns the answer into a tool. EU air freight rates are more volatile and more strongly seasonal than road, on average and significantly, with a clear second-quarter peak, while the country detail shows the pattern is an average rather than a universal rule, with road the riskier leg in the Netherlands and Poland. Adding France reinforced the conclusion rather than weakening it: France is the second most volatile air lane of the six, and the air-over-road volatility ratio widened to 1.63 times. The pipeline is reproducible from open data, the cleaning choices are documented and defensible, and the limits, passenger traffic inside the Eurostat air index, the French freight-only granularity, and the absence of a road EU aggregate, are stated openly.

The Rate Risk Monitor dashboard carries that evidence to the people who act on it. A pricing or procurement manager reads which mode is riskier and where, when in the year to run a tender, and how many euros of within-year swing to budget for, all from a filterable ten-sheet report that is consistent, accessible, and verified across every market. The analysis and the dashboard read from the same fact table, so the story is the same whether a reader meets it as a statistic or as a chart.

For a buyer the practical rule is short. In high-air-risk markets, including Germany, Italy, and France, index or shorten air contracts and time the air tender to the turn-of-year low season, while holding road on longer fixed terms. In the Netherlands and Poland, reverse the treatment, because road carries the larger swing. Size the air budget band by market, from about 3.5 percent in Poland to about 9.6 percent in

Italy. These are tendencies from an official price index, not a single firm's realised cost, but they convert a question usually answered by feel into one answered by evidence.

The recommendations close this report validated rather than asserted. The trough-season air tender paid in 75% of 85 backtested market-years, the volatility gap survives removal of the pandemic years on every window tested, and the Q2 climb repeated out of sample in 10 of 12 market-years in 2024 and 2025. Where the rules break, Poland marginal and France a coin flip, the report says so, because a rule that has not been tried against its failures is not yet a rule.

12. Continuation and enhancements

The dashboard answers the current question well and leaves clear room to grow. A cargo-specific air index, if a paid feed becomes available, would sharpen the air signal beyond the passenger-inclusive proxy, and would also let the five Eurostat countries be measured at the same freight-only level as France, removing the granularity caveat. Extending road past 2024-Q1 as Eurostat publishes would lengthen the comparison. A volume layer would let the dashboard test rate against volume at a lead and a lag. Lane-level rather than country-level data would let a buyer act on a specific trade. A scheduled quarterly refresh from the Eurostat and INSEE sources would keep the dashboard current with no manual step. None of these change the present conclusions. They deepen them.

From the validation chapter, three monitor additions are queued: a hit-rate KPI card on the 05 Timing sheet, a full-window against ex-COVID robustness toggle on the 03 Markets sheet, and a per-market timing verdict chip driven by the backtest table. Each is a small build on the existing model, and each was deferred deliberately so the graded dashboard stays identical to its documented guide.

13. Bibliography

- Eurostat. Services producer price index, quarterly (online data code sts_sepp_q): air transport NACE H51 and road freight transport NACE H49.4, 2021=100, not seasonally adjusted. European Commission, Eurostat dissemination database.
- INSEE. Indices de prix de production des services, transport aerien de fret (CPF 51.21), BDM series 010766683, base 2021, donnees trimestrielles brutes. Institut national de la statistique et des etudes economiques.
- Eurostat. NACE Rev. 2: statistical classification of economic activities in the European Community.
- Levene, H. (1960). Robust tests for equality of variances. In I. Olkin and others (eds.), Contributions to Probability and Statistics. Stanford University Press.
- Brown, M. B., and Forsythe, A. B. (1974). Robust tests for the equality of variances. Journal of the American Statistical Association, 69(346), 364 to 367.
- Kruskal, W. H., and Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. Journal of the American Statistical Association, 47(260), 583 to 621.
- Transport Intelligence and Upply. European Road Freight Rate Benchmark (quarterly industry benchmark on European road freight rates).
- Baltic Exchange. Baltic Air Freight Index, with expanded daily reporting from 2025.
- Trade press coverage of the 2021 to 2022 air-cargo capacity crunch and the 2024 Red Sea disruption.

Appendix

A. Data dictionary

The Eurostat files share the columns below. The INSEE French air file carries a date and a value column with a provenance header.

Column	Meaning
indic_bt	Price indicator, PRC_PRR (producer price) is kept, PRC_PRR_B2B dropped
nace_r2	Activity code, H51 for air, H494 for road
s_adj	Adjustment, NSA (not seasonally adjusted) is kept
unit	Index base, I21 means 2021=100
geo	Country or aggregate code (for example DE, IT, EU27_2020)
TIME_PERIOD	Quarter, for example 2015-Q1
OBS_VALUE	The index level
OBS_FLAG	Eurostat data-quality flag (b break, e estimated, p provisional)

Table 13. Column dictionary for the Eurostat SPPI files.

B. KPI and measure definitions

Measure	Definition
Coefficient of variation	Standard deviation divided by the mean of the level, per mode and country
Rolling volatility	Four-quarter rolling standard deviation of the z-scored level
Quarter-on-quarter change	Period-over-period percent change of the level
Intra-year band	Average within-year range as a percent of the level (air about 6.7, road about 2.8 across the six states)
Air rate risk band	Categorical Low, Medium, or High derived from the air coefficient of variation
Euro exposure (what-if)	Intra-year band times the user's annual air spend
Backtest edge	Trough-season lock (mean of Q1 and Q4) minus peak-season lock (mean of Q2 and Q3), percent of the year mean, per market-year

Table 14. KPI and measure definitions, matching the analysis.

C. Limits, stated once

- The Eurostat air index (NACE H51) includes passenger air transport, so it is a broad air-rate proxy, not a cargo-only price.
- France air is measured at the air-freight level (INSEE CPF 51.21) while the other five use the broader Eurostat H51 air-transport division. A freight-only series tends to swing more, which is part of why France ranks high on air, so France's exact rank should be read with the definition difference in mind.
- Road has no EU or euro-area aggregate, so the comparison and the dashboard run at member-state level on six common countries.

- Country windows differ in start year, so each mode is aligned to its common window per country, ending 2024-Q1, before comparison.
- Findings are tendencies from a price index, not one company's realised costs.

D. Glossary

Term	Meaning
SPPI	Services Producer Price Index, the price providers charge for a service, as an index
NACE	EU activity classification, H51 air transport, H49.4 (H494) road freight
CPF	EU classification of products by activity, 51.21 air freight transport (the INSEE French air level)
NSA	Not seasonally adjusted, seasonality is retained on purpose
z-score	Value minus the series mean divided by its standard deviation, a common scale
Coefficient of variation	Standard deviation divided by the mean, dispersion relative to level
Rolling 4Q std	Standard deviation over a moving four-quarter window, volatility over time
Levene test	Test for equal variance between groups, robust to non-normal data
Kruskal-Wallis	Non-parametric test for whether groups differ, used for the quarter effect

E. Code and reproducibility

The analysis runs from annotated Python scripts. The five-state reference pipeline loads the two Eurostat CSVs, applies the filters and the per-country common window, standardizes the series, computes the volatility metrics, runs the Levene and Kruskal-Wallis tests, builds the seasonal decomposition, and writes the figures and a results file. The six-state extension injects the INSEE French air series, keeps French road from Eurostat, reproduces the five-state numbers exactly, then recomputes on six states. The pipeline uses pandas, numpy, scipy, and statsmodels, with matplotlib for the figures. Every number in this report is reproduced by these scripts and stored in results_6.json. The data is open Eurostat and INSEE data, downloaded without an API key, so the work can be rerun and refreshed each quarter.

The code and supporting files are:

- analysis.py: the five-state reference pipeline, producing the figures and results.json.
- analysis_fr.py: the six-state extension that injects the INSEE French air series, reproduces the five-state numbers, then recomputes on six states and writes results_6.json.
- analysis_upgrades.py: the v2 validation runs (tender-timing backtest, robustness windows, out-of-sample check), writing results_upgrades.json, backtest_air_geo_years.csv and figures 13 to 15. It asserts the stored six-state headline numbers before computing anything new.
- results.json and results_6.json: the machine-readable results (overview, common windows, CV by country, Levene, Kruskal-Wallis, headline) that every number in the report is checked against.
- FR_air_freight_INSEE_010766683.csv: the INSEE French air-freight series with a provenance header.
- fact_freight_6.csv: the six-state long fact table (date, geo, mode, level, z-score, rolling volatility, quarter-on-quarter change, quarter) that feeds the dashboard.
- prices_tidy_6.csv: the six-state tidy long table of levels.
- POWERBI_BUILD_GUIDE.md and RATE_RISK_MONITOR_SPEC.md: the step-by-step build and the DAX measures for the Power BI dashboard.

- Rate Risk Monitor (Power BI project): the graded ten-sheet monitor, star-schema model, published to the Power BI service. Report_2_Dashboard.pbix, the earlier ten-sheet build, is kept as history.
- Report_2_Dashboard.html: the Plotly prototype that opens in any browser.

F. Reproducibility checklist

- Source: Eurostat dataflow sts_sepp_q, air H51 and road H494, downloaded without an API key, plus INSEE BDM series 010766683 (CPF 51.21) for French air.
- Filters: indic_bt = PRC_PRR, unit = I21, s_adj = NSA.
- Scope: six common states DE, IT, ES, NL, PL, FR. France air from INSEE, France road from Eurostat.
- Window: per-country common air-and-road window, ending 2024-Q1.
- The six-state script reproduces the five-state baseline exactly, then recomputes on six states, and rebuilds every figure and number.
- Dashboard: built in Power BI from fact_freight_6.csv, ten sheets, synchronised country slicer, with a five-page PDF kept as a backup.

G. Worked examples

These short worked examples show how the headline numbers are built, so a reader can reproduce the logic by hand.

Coefficient of variation. Take a series with mean level 100 and standard deviation 15. The coefficient of variation is 15 divided by 100, or 0.15. A second series with mean 80 and standard deviation 12 has the same coefficient of variation, 0.15, even though its standard deviation is smaller, which is why dividing by the mean is the fair comparison across countries at different levels.

Z-score. For the same first series, a quarter at 130 has a z-score of (130 minus 100) divided by 15, or +2.0, two standard deviations above the series mean. A quarter at 92.5 has a z-score of -0.5. Expressing every series in z-scores lets the German and Italian air lines share one vertical axis.

Quarter-on-quarter change. If the air index reads 110 in one quarter and 113.3 the next, the change is 113.3 divided by 110 minus 1, or about +3.0 percent. The seasonality test groups these percentage changes by calendar quarter and asks whether the four groups differ.

Reading the variance test. The Levene statistic of 65.0 with p about 3e-15 says the spread of air changes and the spread of road changes are very unlikely to be equal by chance. The effect size is separate: the air change standard deviation of 0.053 against road's 0.016 is the 3.4 times figure quoted in the headline.

Reading the seasonality test. The Kruskal-Wallis statistic of 49.6 with p about 1e-10 for air says the quarter-on-quarter change is not the same across the four quarters. The direction is read off the quarter means, which peak in the second quarter at about +3.2 percent for air.

Quantity	How it is computed	Six-state value
Mean air CV	mean over countries of (std / mean of level)	0.155
Mean road CV	mean over countries of (std / mean of level)	0.095
Air-over-road CV ratio	mean air CV divided by mean road CV	1.63
Air vs road change spread	std of air QoQ divided by std of road QoQ	about 3.4
Intra-year air band	average within-year range as percent of level	6.7 percent

Table 15. Worked summary of the headline quantities.